PART I · WHICH NOTE

Pitch

Every note is higher or lower than another. This part starts with the keyboard you can see and touch, moves to the staff where those keys are written, and ends with the families of notes (scales and keys) that whole pieces are built from.

Keyboard high and low, the patterns › **Staff** lines, spaces and clefs › **Reading the notes** every note on the staff › **Sharps and flats** the black keys › **Intervals** the distance between notes › **Scales and keys** notes in families

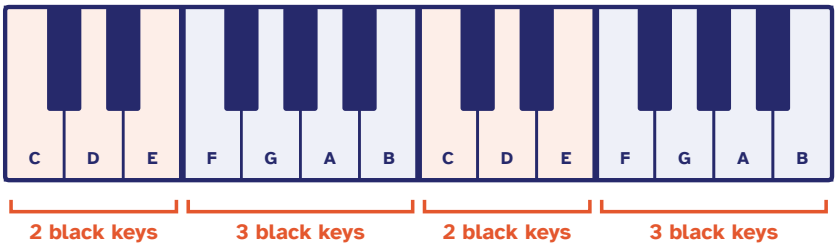
IN THIS PART

The keyboard 22
The staff and the clefs 24
Reading the notes 26
The staff and the keyboard 28
Sharps, flats and naturals 30
Counting intervals 32
Major, minor and perfect 34
Major scales 36
Minor scales 38
Keys and key signatures 40
Finding the key 42
The circle of fifths 44
Tips: when a note sounds wrong 46

PART I · PITCH · 1

The keyboard

Keys to the **left** sound **lower**; keys to the **right** sound **higher**. The black keys come in groups of **two** and **three**, and that pattern repeats all the way along: it is how you find every note.



The white keys

Seven letters, **A B C D E F G**, then the letters start again, like the days of the week. **C D E** sit around each group of two black keys (C just left of it), **F G A B** around each group of three (F just left of it): the two tints on the picture.

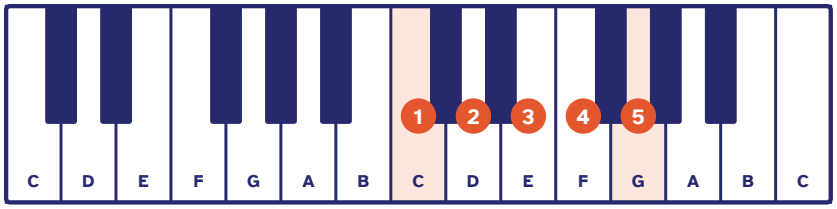
Middle C and octaves

Middle C is the C nearest the middle of the keyboard. From one C to the next is an **octave**: eight white keys, counting both. Octaves are numbered from the bottom: middle C is **C4**, the C below it C3, the C above C5; every key up to the next B shares the number (D4, E4 … B4).

TERM	MEANING	FIRST MET
pitch	how high or low a sound is	Level 0
register	a part of the keyboard: low, middle or high	Level 0
semitone	from one key to the very next key, black or white	Level 0
tone	two semitones	Level 0
octave	from one note to the next with the same name (8 white keys, 12 semitones)	Level 0

IN ACTION

Find G4 on the keyboard



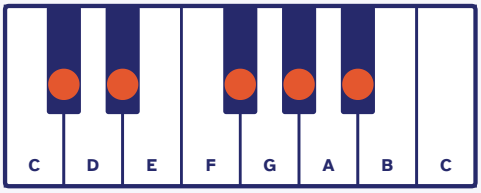
1. **Find middle C (C4):** the C just left of the two black keys nearest the middle.
2. **Count up the letters**, one white key each: C D E F G is five keys.
3. **Same octave**, so the number stays 4.

The key: **G4**. One octave higher, the next G, is G5.

PART I · PITCH · 1 · CONTINUED

YOU KNOW THIS

Amazing Grace, on the black keys only

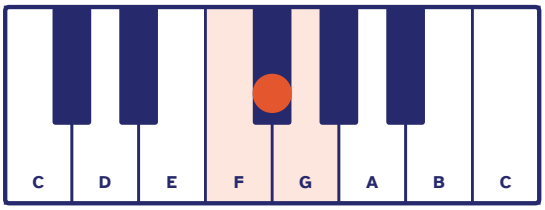


The whole tune fits on the **five black keys**: D♭ G♭ B♭ G♭ B♭ A♭ G♭ E♭ D♭. The groups of two and three are a little melody machine.

Try it. Start on the lower key of a pair of black keys and find the tune by ear.

IN ACTION

Find F♯4



1. **Find F4:** the white key just left of the group of three black keys above middle C.
2. **Sharp means up:** move one key to the right.
3. **The very next key** is black: that is F♯.

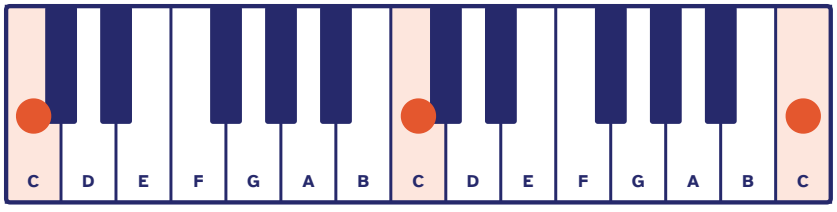
F♯4 is the **black key** just right of F4, the first of the group of three.

WATCH OUT

- × Counting octaves from the left end of the piano: octave numbers always change at C, not A.
- × Calling every black key a sharp: each one has a flat name too (p. 30).

IN ACTION

Find C3, C4 and C5



1. **Find middle C (C4):** just left of the two black keys in the middle.
2. **Count seven white keys down:** C3.
3. **Count seven white keys up:** C5.

Every C sits just left of a group of **two** black keys, an octave apart.

SEE ALSO

The staff and the clefs p. 24 · The staff and the keyboard p. 28

PRACTISE IT

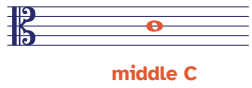
Workbook Pitch 1 · Online: Library › Keyboard

PART I · PITCH · 2

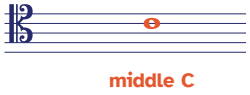
The staff and the clefs

Music is written on a **staff** (or stave) of five lines and four spaces. Lines and spaces are counted from the **bottom**. Higher notes sit higher on the staff. A **clef** at the start tells you which notes the lines and spaces stand for.

CLEF	NAME	HOW IT SHOWS ITS NOTE	FIRST MET
𝄞	**treble clef** (G clef)	curls round the line for G (line 2)	Level 0
𝄢	**bass clef** (F clef)	its two dots sit either side of the line for F (line 4)	Level 0



Alto clef: middle C on line 3. Used by the viola.

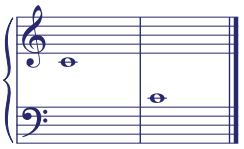


Tenor clef: middle C on line 4. Cello, bassoon and trombone, up high.

Other clefs. C clefs: the line through the centre of the clef is middle C. Piano music never uses them, but pianists meet them in theory exams and when reading the viola or cello part in chamber music or a score (Level 2).

The grand staff

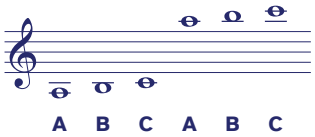
Piano music uses two staves joined by a brace: the treble staff, usually for the right hand, above the bass staff, usually for the left. **Middle C** sits between them on a short line of its own, and can be written in either staff.



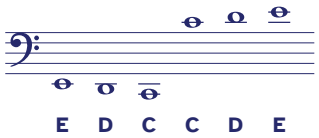
Middle C in the treble staff (left) and in the bass staff (right): the same key.

Ledger lines

Notes above or below the staff sit on short extra lines called **ledger lines**, spaced like the staff lines.



Treble: A, B, C below; A, B, C above.



Bass: E, D, C below; C, D, E above.

Octave signs

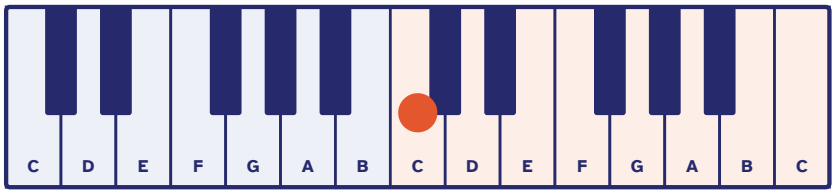
SIGN	MEANING
8va	an octave higher, as far as the dotted line
8vb	an octave lower
15ma	two octaves higher (Level 4)
loco	back where it is written

First met: Level ½. Octave signs save writing many ledger lines.

PART I · PITCH · 2 · CONTINUED

IN ACTION

Which staff does this note belong to?
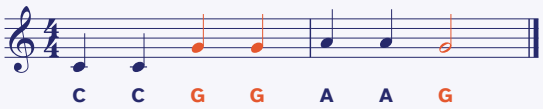


1. **Below middle C** (blue): the bass staff, usually the left hand.
2. **Above middle C** (pink): the treble staff, usually the right hand.
3. **Middle C itself** (the red dot): either staff, whichever hand plays it.

Low keys: **bass clef**. High keys: **treble clef**. The two staves meet at middle C.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



The treble clef curls round **line 2: G**, the very note Twinkle leaps up to. Middle C hangs on its own ledger line below.

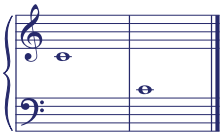
Try it. Point to the G line, then play C C G G A A G.

WATCH OUT

- × Reading bass-clef notes as if they were treble: check the clef on every line.
- × Thinking middle C has two keys because it can be written in two places.

IN ACTION

Middle C in both clefs



1. **Treble staff:** middle C sits on a ledger line below.
2. **Bass staff:** middle C sits on a ledger line above.
3. **On the keyboard:** both are the same key.

One key, two ways to write it: the hand that plays it chooses the staff.

SEE ALSO

Reading the notes p. 26 · The staff and the keyboard p. 28

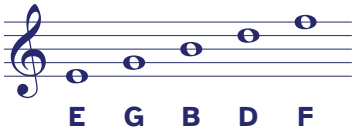
PRACTISE IT

Workbook Pitch 2 · Online: Library › Staff and clefs

PART I · PITCH · 3

Reading the notes

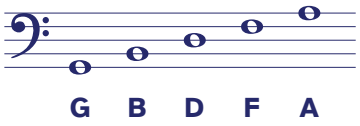
Each line and space has one letter name. Learn them in two ways: by the patterns below, and by a few **landmark notes** you can find instantly and count from.



Treble clef: the lines



Treble clef: the spaces



Bass clef: the lines



Bass clef: the spaces

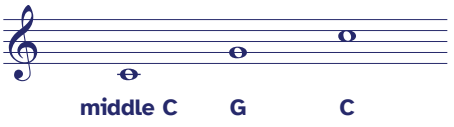
STAFF	NOTES, BOTTOM TO TOP	TO REMEMBER
Treble lines	E G B D F	*Every Good Boy Deserves Fun*
Treble spaces	F A C E	spells *FACE*
Bass lines	G B D F A	*Great Big Dragon Flies Away*
Bass spaces	A C E G	*All Cows Eat Grass*

Landmark notes

Find these first. Every other note is a **step** (line to space) or a **skip** (line to line) away.
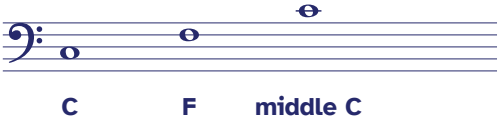
Treble clef

Middle C on its ledger line below; the G the clef curls round (line 2); the C in the third space.



Bass clef

The C in the second space; the F between the clef's dots (line 4); middle C on its ledger line above.



THINK OF IT Landmark notes are like lamp posts on a dark road: walk from one to the next and you are never lost.

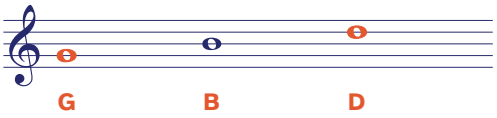
WATCH OUT

- × Naming bass-clef notes with treble names: in the bass, the bottom line is G, not E.
- × Counting ledger lines from the wrong end: start from the staff and count outwards.

PART I · PITCH · 3 · CONTINUED

IN ACTION

Read a note you are not sure of
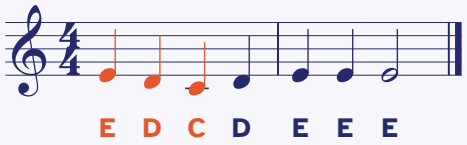


1. **The clef:** treble.
2. **A landmark:** the G the clef curls round, on line 2.
3. **Skip up** line to line, one letter jumped each time: G, B, D.

The note on line 4 is **D**.

YOU KNOW THIS

Mary Had a Little Lamb

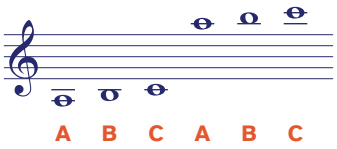


E D C: **line, space, ledger line**, one step down each time. Read the shape and the tune walks down with it.

Try it. Read the notes aloud before you play them.

IN ACTION

Read a ledger-line note



1. **Start at the staff edge:** the top line of the treble staff is F.
2. **Step up** one note at a time: G (space), A (first ledger line).
3. **Below the staff** count down the same way: D, C, B, A.

Ledger lines are the staff carried on: count by steps from the nearest line.

WATCH OUT

- × Reading the bass staff with treble names: check the clef first, every time.
- × Counting lines from the top: lines and spaces are always counted from the bottom.

SEE ALSO

The staff and the clefs p. 24 · The staff and the keyboard p. 28

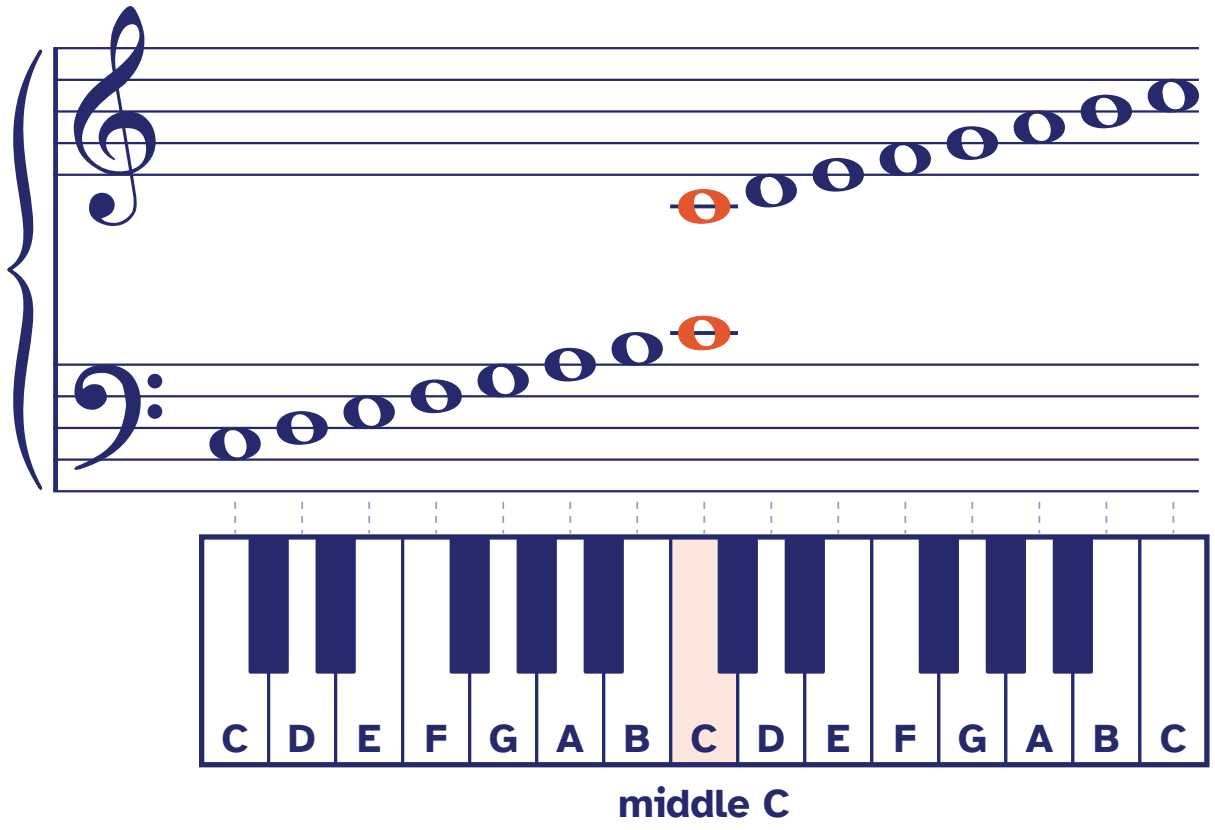
PRACTISE IT

Workbook Pitch 2 · Online: Library › Staff and clefs

PART I · PITCH · 4

The staff and the keyboard

Every line and space of the grand staff is one white key. Read from left to right, the notes climb the staff exactly as the keys climb the keyboard: two octaves from C3 to C5, with middle C in red where the two staves meet.



What the map shows

Each step up the staff (line to space, space to line) is the next **white** key. The black keys have no line or space of their own: they are written with a sharp or a flat on a white key's line or space (p. 30).

Two ways to write one key

Middle C can sit on a ledger line below the treble staff or above the bass staff. Both mean the same key; the music chooses the staff of the hand that plays it.

THINK OF IT The staff is a ladder laid on its side beside the keyboard: each rung, a line or a space, is the next white key.

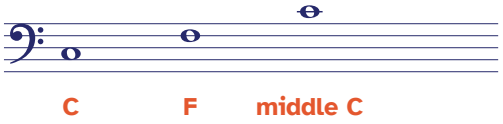
WATCH OUT

× Thinking the black keys have their own lines: they borrow a white key's line or space with a sharp or flat.

PART I · PITCH · 4 · CONTINUED

IN ACTION

From the page to the key

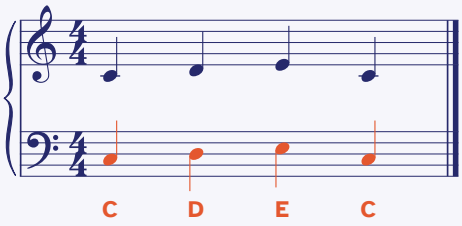


1. **Name the note on the staff:** line 4 of the bass staff, between the clef's dots, is F.
2. **Count down from middle C:** C B A G F, the fifth white key down.
3. **Play it** with the left hand: that is F3.

The bass-clef F is **F3**, five white keys below middle C.

YOU KNOW THIS

Frère Jacques, in both hands



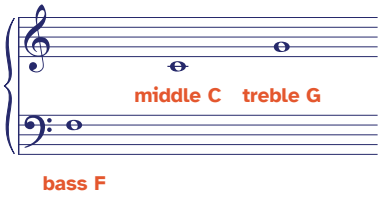
The left hand plays the **same letters**, C D E C, one octave lower, written in the bass clef.

Try it. Play it right hand, then left hand, then both together.

THINK OF IT Reading music is like reading a map while driving: glance at the page, trust your hands to know the distances.

IN ACTION

Landmarks either side of middle C



1. **Find middle C** on the map: between the two staves.
2. **Treble G** is five white keys above it.
3. **Bass F** is five white keys below it.

The landmarks sit at the same distance either side of middle C.

SEE ALSO
The keyboard p. 22 · Reading the notes p. 26

PRACTISE IT
Workbook Pitch 1–2 · Online: Library › Staff and clefs
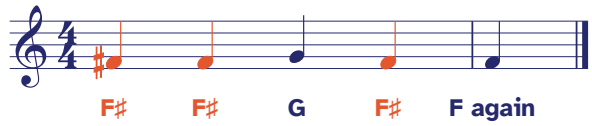
PART I · PITCH · 5

Sharps, flats and naturals

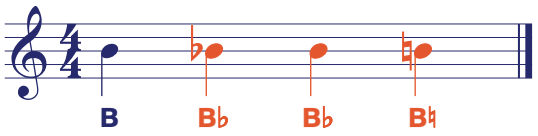
An **accidental** written before a note changes its pitch, usually by a semitone. On the keyboard, a sharp usually means the black key to the right, a flat the black key to the left.

SIGN	NAME	WHAT IT DOES	FIRST MET
♯	sharp	raises a note by a semitone	Level 0
♭	flat	lowers a note by a semitone	Level 0
♮	natural	cancels a sharp or flat	Level ½
𝄪	double sharp	raises by two semitones	Level 4
𝄫	double flat	lowers by two semitones	Level 4

An accidental lasts for the bar



The sharp still counts later in the bar; after the bar line, F is F again.



A natural cancels the flat for the rest of the bar.

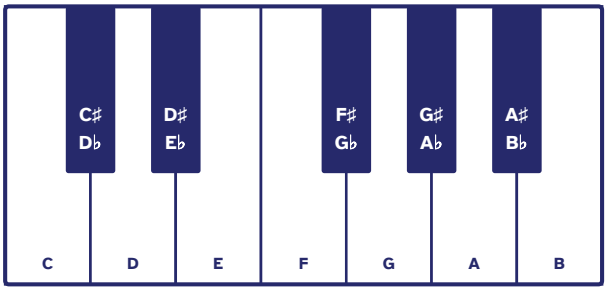
Only that line or space. A sharp on the F in the first space does not change the F an octave higher.

A tie carries it on. A note tied across the bar line keeps its accidental.

Reminders. A courtesy accidental, sometimes in brackets, just reminds you after a bar line.

One key, two names

Every black key has two names: the key between F and G is both **F♯** and **G♭**. Notes that sound the same but are written differently are **enharmonic**. White keys can have sharp or flat names too: **E♯** is F, **C♭** is B.
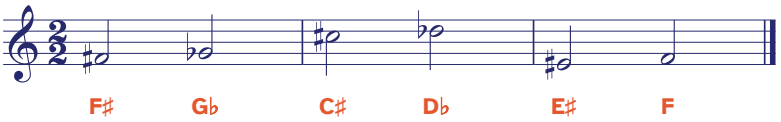


THINK OF IT A key signature is the principal's notice on the school board: it speaks to every student, every day. An accidental is a teacher stopping at one student's desk: the message is for that note only, and it lasts until the next bar line.

PART I · PITCH · 5 · CONTINUED

IN ACTION

One key, two names
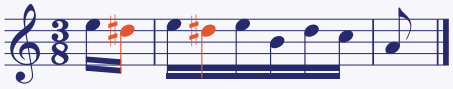


1. **The first bar:** F♯, then G♭.
2. **Find them:** both are the black key between F and G.
3. **The last bar:** no black key sits above E, so E♯ is the white key F.

Same key, different names: these are **enharmonic** notes.

YOU KNOW THIS

Für Elise (Beethoven)



E, **D♯**, E, **D♯**: a semitone down and back. The sharp is written again in the new bar, because the bar line wiped it out.
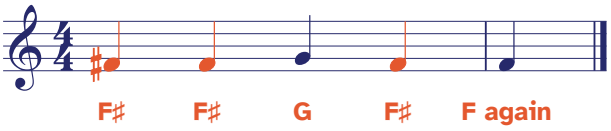
Try it. Play E and D♯ back and forth: the black key just left of E.

WATCH OUT

- × Forgetting that an accidental lasts until the bar line.
- × Carrying an accidental into the next bar: after the bar line, the key signature rules again.

IN ACTION

How long does the sharp last?



1. **The first note:** F♯.
2. **Later in the bar:** F is still F♯.
3. **After the bar line:** F is plain F again.

An accidental lasts **until the bar line**, then the key signature rules again.

SEE ALSO

Keys and key signatures p. 40

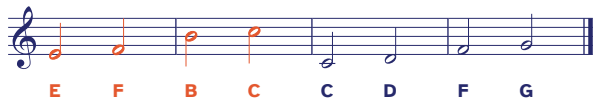
PRACTISE IT

Workbook Pitch 3 · Online: Library › Signs

PART I · PITCH · 6

Counting intervals

An **interval** is the distance between two notes. A **semitone** is the very next key; a **tone** is two semitones. An interval's **number** comes from counting letter names, both ends included.

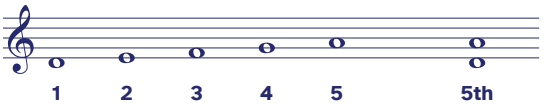


E–F and B–C are semitones; C–D and F–G are tones.



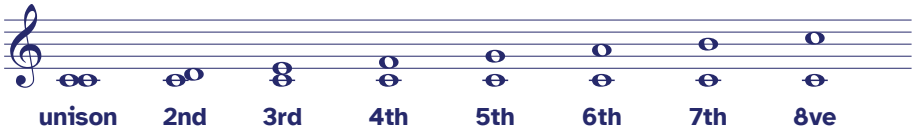
Melodic: one note after the other. Harmonic: together.

Counting the number



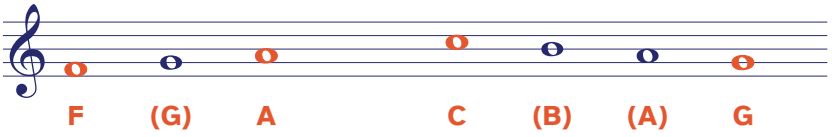
D up to A: D E F G A is five letters, so the interval is a 5th, whether the notes come one after the other or together.

Interval numbers from C



IN ACTION

How far apart are these notes?



1. **Call the first note 1:** F is 1.
2. **Count every letter to the second note:** F G A is 3 letters: a 3rd.
3. **Going down works the same:** C B A G is 4 letters: a 4th.
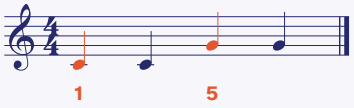
F up to A is a **3rd**; C down to G is a **4th**. Sharps and flats never change the number.

THINK OF IT Put your thumb on C and your little finger on G: five fingers on five white keys, a 5th. Your hand measures intervals for you.

PART I · PITCH · 6 · CONTINUED

YOU KNOW THIS

Twinkle, Twinkle, Little Star

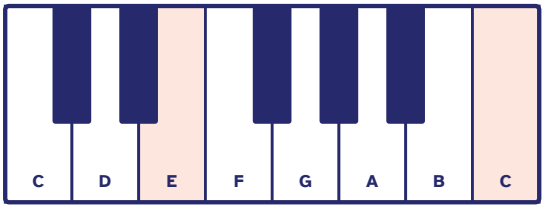


The leap from “twin-” to “kle” is C up to G: C D E F G, **five letters, a 5th**.

Try it. Sing the first two notes of any song, then count the letters between them.

IN ACTION

Name an interval on the keyboard



1. **Find both notes:** E and the C above it.
2. **Count the letters:** E F G A B C is six.
3. **Both ends count:** so the interval is a 6th.

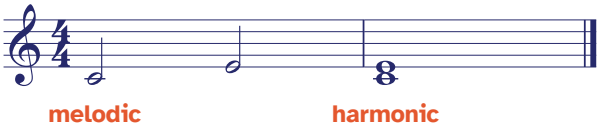
E up to C is a **6th**.

WATCH OUT

- × Counting only the gaps, not the notes: always count both ends.
- × Letting a sharp or flat change the number: C to G♯ is still a 5th.

IN ACTION

Melodic or harmonic?



1. **The first bar:** C then E, one after the other.
2. **The second bar:** C and E together.
3. **Count both:** C D E, a 3rd each time.

The same 3rd: **melodic** when played one after the other, **harmonic** when played together.

SEE ALSO

Major, minor and perfect p. 34 · Major scales p. 36

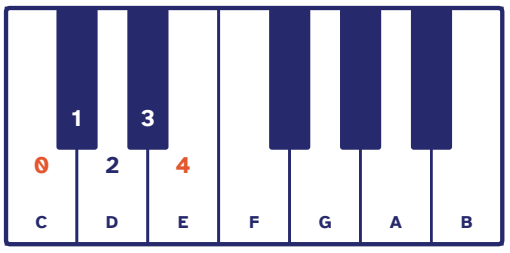
PRACTISE IT

Workbook Pitch 4 · Online: Library › Intervals

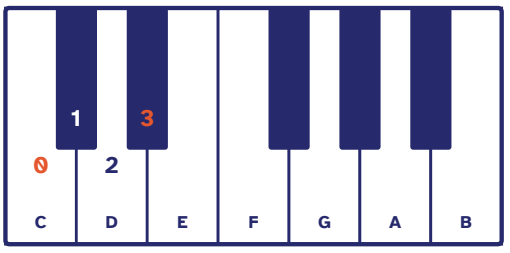
PART I · PITCH · 7

Major, minor and perfect
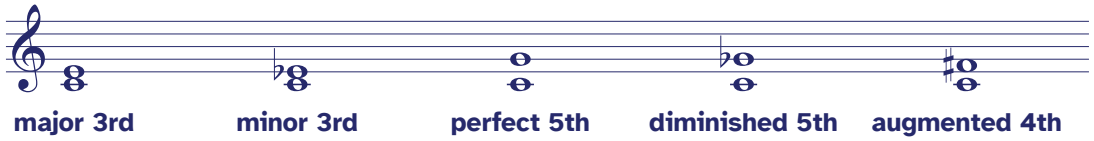
Two intervals with the same number can differ in size. C–E and C–E♭ are both 3rds, but one is a semitone wider. The **quality** (major, minor, perfect…) comes from counting **semitones**.



C to E: count every key, black and white. 4 semitones: a major 3rd.



C to E♭: 3 semitones: a minor 3rd.



Size in semitones

NUMBER	SMALLER	MAIN	WIDER	FIRST MET
unison		0 (perfect)		Level 1
2nd	1 (minor)	2 (major)		Level 1
3rd	3 (minor)	4 (major)		Level ½
4th		5 (perfect)	6 (augmented)	Level 1
5th	6 (diminished)	7 (perfect)		Level 1
6th	8 (minor)	9 (major)		Level 2
7th	10 (minor)	11 (major)		Level 2
octave		12 (perfect)		Level 1

2nds, 3rds, 6ths, 7ths
diminished ‹ **minor** ‹ **major** ‹ augmented

unisons, 4ths, 5ths, octaves
diminished ‹ **perfect** ‹ augmented

Each step on the ladder is one semitone. Six semitones (an augmented 4th or diminished 5th) is the tritone. Turned upside down, an interval's numbers add to 9 and major and minor swap: a major 3rd becomes a minor 6th.

THINK OF IT A major and a minor 3rd are a smile and a frown: the same face, one small change at the corners.

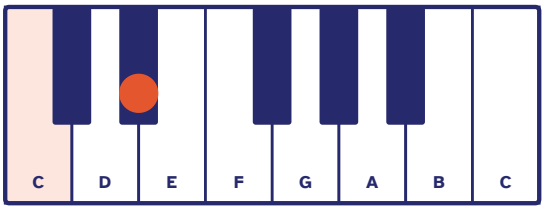
WATCH OUT

- × Counting white keys only: count every key, black and white.
- × Calling a 4th or 5th "major": they are perfect.

PART I · PITCH · 7 · CONTINUED

IN ACTION

Major or minor 3rd?
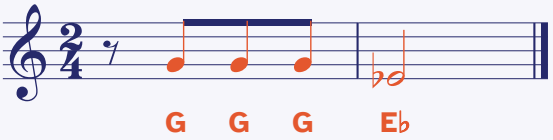


1. **Count the letters:** C D E is three, so it is a 3rd.
2. **Count the semitones:** C to C♯, D, E♭ is three.
3. **Three semitones** make a minor 3rd; four would be major.

C to E♭: a **minor 3rd**.

YOU KNOW THIS

Beethoven, Symphony No. 5

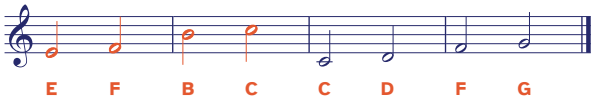


Da-da-da-DUM: G down to **E♭** is four semitones, a **major 3rd**. Three semitones would be minor.

Try it. Count the keys from G down to E♭: F♯, F, E, E♭.

IN ACTION

Tone or semitone?



1. **E to F:** the very next key, no black key between: a semitone.
2. **C to D:** one key between (C♯): a tone.
3. **B to C** and **F to G**: check the same way.

E–F and B–C are semitones; C–D and F–G are tones.

SEE ALSO

Counting intervals p. 32 · Chords: Harmony (later)
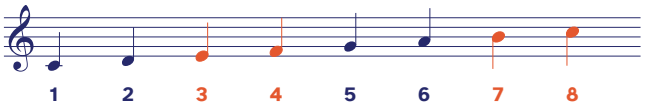
PRACTISE IT

Workbook Pitch 4 · Online: Library › Intervals

PART I · PITCH · 8

Major scales

A **scale** is a ladder of notes from one key note up to the next, in a fixed pattern of tones (T) and semitones (S). The pattern, not the starting note, makes a scale major or minor.



C major: T T S T T T S. The semitones fall between 3–4 and 7–8.

SCALE	STEPS	FIRST MET
major	T T S T T T S	Level ½
natural minor	T S T T S T T	Level ½
harmonic minor	T S T T S T+S S	Level ½
melodic minor (up)	T S T T T T S; down as the natural minor	Level 2
chromatic	every semitone	Level 2
pentatonic (black keys)	T T T+S T T+S	Level 0

Degree names

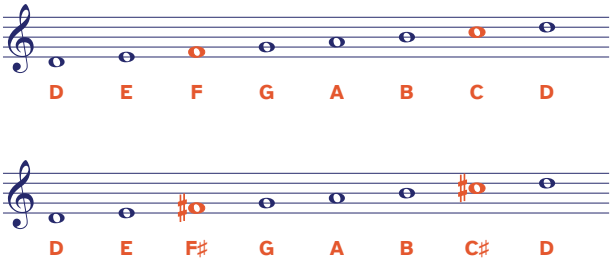
1 tonic · **2** supertonic · **3** mediant · **4** subdominant · **5** dominant · **6** submediant · **7** leading note

Major or minor?

The same key note can start a major or a minor scale: C major and C minor. The pattern decides. Major sounds bright and settled; minor, darker.

IN ACTION

Build D major



1. **Write the eight letters** from D to D, every letter once.
2. **Check the steps** against T T S T T T S: E–F is too small, B–C is too small.
3. **Raise F and C** a semitone to fix them.
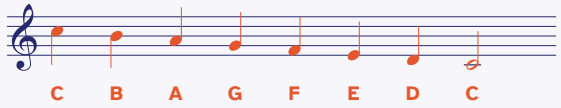
D major: **D E F♯ G A B C♯ D**, two sharps.

THINK OF IT A scale is a staircase with two short steps. Wherever you build it, the short steps sit in the same places: between 3 and 4, and between 7 and 8.

PART I · PITCH · 8 · CONTINUED

YOU KNOW THIS

Joy to the World



The first line is the **C major scale**, top to bottom: C B A G F E D C.

Try it. Play C major down from the top C and sing "Joy to the world" over it.

IN ACTION

Build F major

1. **Write the letters** from F to F: F G A B C D E F.
2. **Check the steps** against T T S T T T S: A to B is a tone, but it must be a semitone.
3. **Lower B** to B♭: now A to B♭ is a semitone and B♭ to C a tone.

F major: **F G A B♭ C D E F**, one flat.

WATCH OUT

- × Using the same letter twice: every letter appears once (F♯, not G♭, in G major).
- × Forgetting the semitone between 7 and 8: the leading note pulls up to the key note.
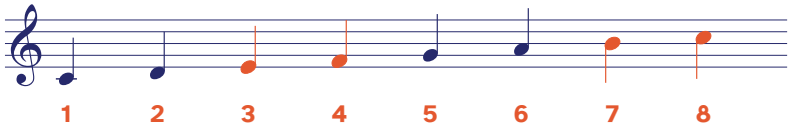
HOW TO

Practise a scale

1. **Hands separately,** slowly, saying the finger numbers aloud.
2. **Fingering for C, G, D, A and E major** (right hand): 1 2 3, thumb under, 1 2 3 4 5.
3. **Listen for evenness:** every note the same length and loudness.
4. **Then hands together,** one octave, then two.

IN ACTION

Where are the semitones?



1. **Number the degrees** 1 to 8 from C.
2. **Look for no black key between:** E–F and B–C.
3. **Those are degrees 3–4 and 7–8.**

Every major scale has its semitones between **3–4** and **7–8**.

SEE ALSO

Minor scales p. 38 · Keys and key signatures p. 40

PRACTISE IT
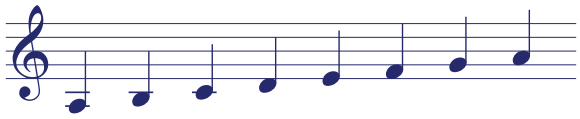
Workbook Pitch 5 · Online: Scale finder, Library › Scales

PART I · PITCH · 9
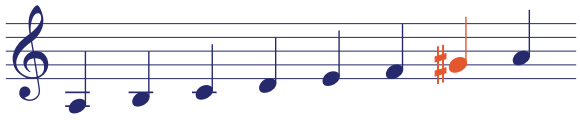
Minor scales

Every minor scale starts from the natural minor, the notes of its relative major from the 6th degree up. Two small changes give the harmonic and melodic forms, which sound stronger at the top.

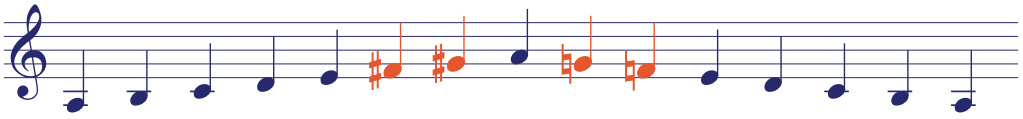
A minor in three forms



Natural: the notes of C major, from A to A. T S T T S T T.



Harmonic: the 7th raised a semitone (**G♯**). The gap F–G♯ is a tone and a half.



Melodic: the 6th and 7th raised going up (**F♯ G♯**), lowered coming down (**G♮ F♮**).

Relative minor

Same key signature, three semitones below the major: C major and A minor share every note.

Parallel minor

Same key note, different signature: C major (no sharps or flats) and C minor (three flats).

THINK OF IT The relative minor is the same clothes worn by a different person: the same notes, a new key note. The parallel minor is the same person in different clothes: the same key note, some notes changed.

IN ACTION

Build E harmonic minor
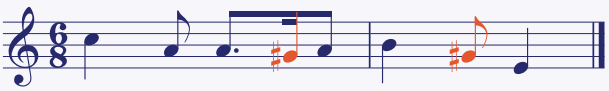
1. **Borrow G major's notes** starting on E: E F♯ G A B C D E.
2. **Raise the 7th:** D becomes D♯.
3. **Check the gap** C to D♯: a tone and a half.

E harmonic minor: **E F♯ G A B C D♯ E**.

PART I · PITCH · 9 · CONTINUED

YOU KNOW THIS

Greensleeves

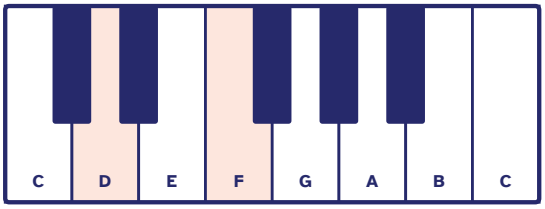


A minor, with a G♯: the raised 7th of the harmonic minor, leaning hard up to A.

Try it. Play it once with G♯, once with G: hear the pull disappear.

IN ACTION

Find the relative minor of F major



1. **Start on F,** the key note.
2. **Count three semitones down:** E, E♭, D.
3. **Check the signature:** D minor has one flat, B♭, like F major.
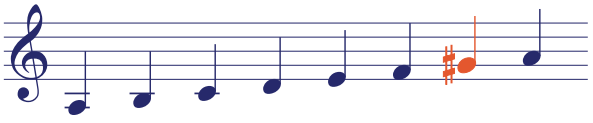
The relative minor of F major is **D minor**.

WATCH OUT

- × Raising the 7th in the key signature: it is written as an accidental each time.
- × Playing the melodic minor's raised notes on the way down.

IN ACTION

Natural or harmonic minor?



1. **Start on A** and read up to A.
2. **Look at the 7th:** G♯, raised.
3. **Nothing else changes** from the natural minor.

A raised 7th only: the **harmonic minor**.

SEE ALSO

Major scales p. 36 · Keys and key signatures p. 40

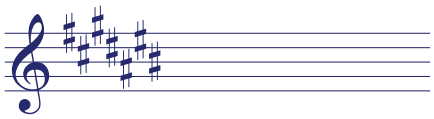
PRACTISE IT

Workbook Pitch 5 · Online: Scale finder, Library › Scales

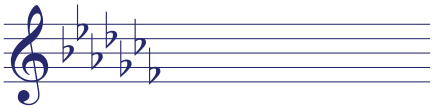
PART I · PITCH · 10

Keys and key signatures

A **key signature** at the start of each line lists the sharps or flats the piece uses throughout, so they do not have to be written before every note. They always appear in the same order.



Sharps: F C G D A E B. The key note is a semitone above the last sharp.

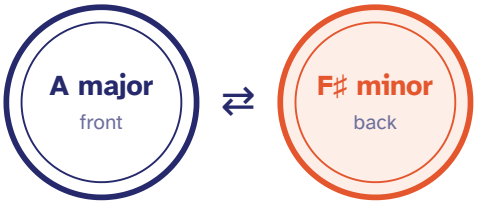


Flats: B E A D G C F. From two flats, the second-last flat names the key.

To remember the order: **F**ather **C**harles **G**oes **D**own **A**nd **E**nds **B**attle; backwards for the flats: **B**attle **E**nds **A**nd **D**own **G**oes **C**harles' **F**ather.

MAJOR	RELATIVE MINOR	SIGNATURE	FIRST MET
C major	A minor	—	Level ½
G major	E minor	1♯	Level ½
D major	B minor	2♯	Level 1
A major	F♯ minor	3♯	Level 3
E major	C♯ minor	4♯	Level 4
B major	G♯ minor	5♯	Level 4
F major	D minor	1♭	Level ½
B♭ major	G minor	2♭	Level 1
E♭ major	C minor	3♭	Level 3
A♭ major	F minor	4♭	Level 4
D♭ major	B♭ minor	5♭	Level 4

Six and seven sharps or flats (F♯, C♯, G♭, C♭ major and their minors) come at Levels 4–6.



One signature, two keys

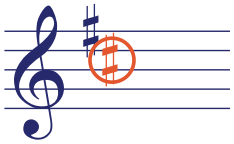
Every key signature belongs to **one major key and one minor key**, like the two sides of a coin: three sharps are A major or F♯ minor. The minor is the **relative minor**, three semitones below the major.

THINK OF IT The key signature is the notice board for the whole piece: every F, in every octave, on every line, follows it. Only an accidental, a word to one note, can overrule it, and only until the bar line.

PART I · PITCH · 10 · CONTINUED

IN ACTION

Name the keys
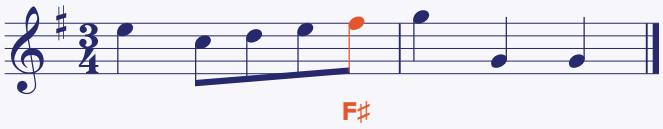


1. **Count:** two sharps, F♯ and C♯.
2. **The last sharp** is C♯; one semitone up is D.
3. **The relative minor,** three semitones below D, is B.

Two sharps: **D major** or **B minor**.

YOU KNOW THIS

Minuet in G



One sharp in the key signature: every F is **F♯**, like the one just before the top G. That sharp is why it is in G major.
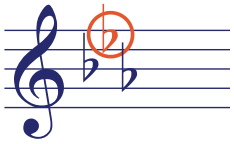
Try it. Find every F in the piece and play it as F♯.

WATCH OUT

- × Writing a key signature in the wrong order: sharps always F C G D A E B.
- × Forgetting that a signature names two keys: a major and its relative minor.

IN ACTION

Read a flat key signature



1. **Count the flats:** three, B♭ E♭ A♭.
2. **The second-last flat** is E♭: the key note.
3. **Its relative minor,** three semitones down, is C minor.

Three flats: **E♭ major** or **C minor**.

SEE ALSO

Major scales p. 36 · Finding the key p. 42 · The circle of fifths p. 44

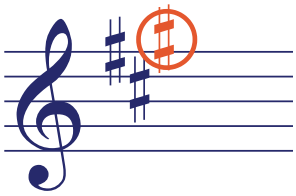
PRACTISE IT

Workbook Pitch 5 · Online: Scale finder, Library › Keys
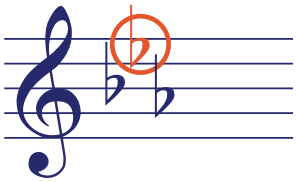
PART I · PITCH · 11

Finding the key

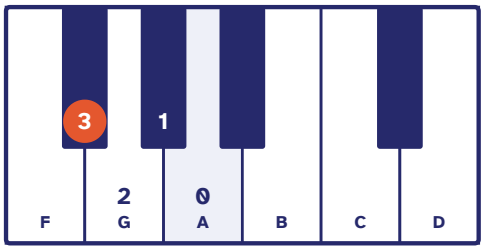
A key signature names two keys, a major and its relative minor. Two quick rules (and one to remember) find the major; one more finds the minor; then the music itself tells you which of the two it is.



Sharps: the last sharp, then one semitone up.
Last sharp **G♯** → up a semitone → **A major**.



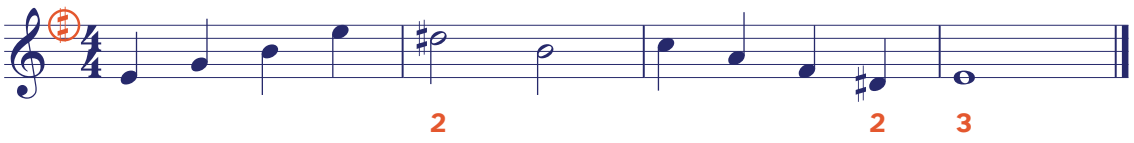
Flats: the second-last flat is the key.
B♭ **E♭** A♭ → **E♭ major**. One flat alone is F major.



Major to minor: three semitones down.
A, G♯, G, **F♯** → **F♯ minor**, the relative minor of A major (count every key, black and white).

IN ACTION

What key is this tune in?
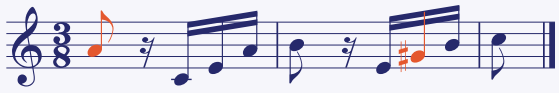


1. **The key signature:** one sharp, so G major or its relative, E minor.
2. **The clue notes:** D♯ keeps appearing. D♯ is the raised 7th of E minor; G major has no D♯.
3. **The last note:** E, the key note of E minor.

The key: **E minor**. Check it the same way with any piece you play.

PART I · PITCH · 11 · CONTINUED

YOU KNOW THIS

Für Elise (Beethoven)



No key signature, a G♯ as the clue, and the music keeps coming home to A: **A minor**.

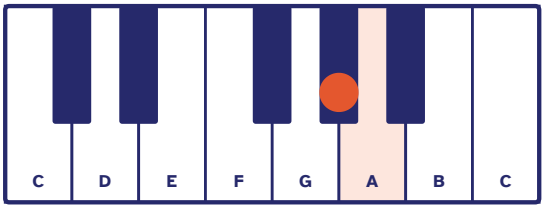
Try it. Play A, then G♯ A: hear the G♯ lean into home.

Minor keys and their clue notes

MINOR KEY	SIGNATURE	CLUE NOTE (RAISED 7TH)
A minor	none	G♯
E minor	1♯	D♯
D minor	1♭	C♯
B minor	2♯	A♯

IN ACTION

Major or minor? No sharps or flats



1. **The signature:** none, so C major or A minor.
2. **The clue notes:** G♯ appears. G♯ is the raised 7th of A minor.
3. **The last note:** A.

No sharps or flats, G♯ clues, ending on A: **A minor**.

IN ACTION

Read a sharp key signature



1. **Count the sharps:** one, F♯.
2. **The last sharp** is F♯; one semitone up is G.
3. **Its relative minor,** three semitones below G, is E.

One sharp: **G major** or **E minor**.

SEE ALSO

Keys and key signatures p. 40 · The circle of fifths p. 44

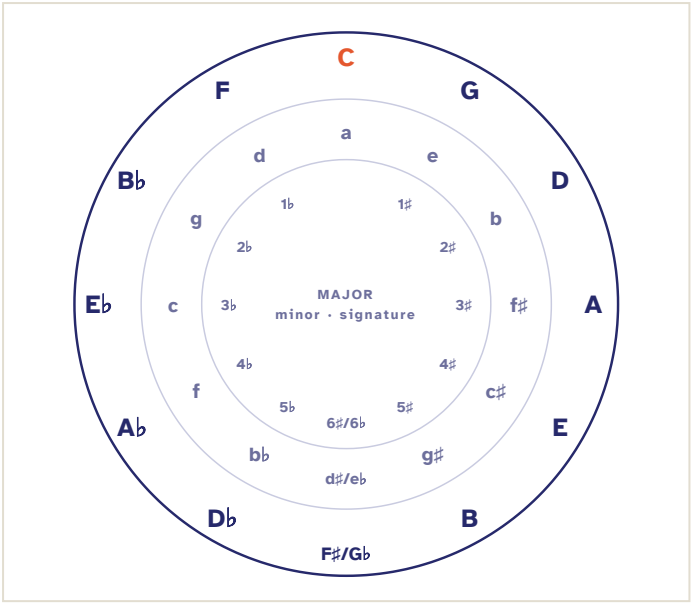
PRACTISE IT

Workbook Pitch 5 · Online: Scale finder, Library › Keys

PART I · PITCH · 12

The circle of fifths

All the keys on one clock face. Going round clockwise, each key is a **5th higher** than the last and has **one more sharp** (or one fewer flat). Anticlockwise, each is a 4th higher, a 5th lower, with one more flat.



Reading the circle

Outer ring: major keys. **Middle ring:** their relative minors (small letters). **Inner ring:** the key signature.

Neighbours are close

Keys next to each other share all but one note, so music moves between them easily: from C major to G major or F major, or to the relative minor. These are the **related keys**.

Six ways to use it

Key signatures
Count the steps from C: clockwise for sharps, anticlockwise for flats.

Relative minors
Look straight in from any major key.

The main chords
A key's two neighbours give its other main chords: in C, the chords of F and G.

Cadences
The strongest close, V to I, is one step anticlockwise: G to C.

Changing key
Pieces usually move one step round, or to the relative minor.

Scale practice
Go round the circle: one new sharp or flat each time.

IN ACTION

Why do G major and D major sound so close?

1. **Find both:** D is one step clockwise from G.
2. **Compare the scales:** G major has C; D major has C♯.
3. **Count:** six notes the same, one different.
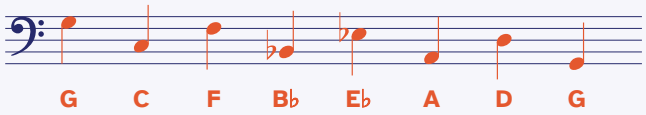
Neighbours share six of their seven notes, so music moves between them easily.

THINK OF IT Like a colour wheel: neighbours blend, opposites clash. C and G sit side by side; C and F♯ sit opposite.

PART I · PITCH · 12 · CONTINUED

YOU KNOW THIS

Passacaglia (Handel), the bass



Each bass note is a **5th below** the one before (or a 4th above): G C F B♭ E♭ A D G, one step round the circle each chord.

Try it. Play the bass line slowly and hear it pull towards the last G.

IN ACTION

Find A major's key signature

1. **Start at C** at the top of the circle: no sharps.
2. **Go clockwise:** G, D, A is three steps.
3. **Three steps,** three sharps: F♯, C♯, G♯.

A major has **three sharps**.

WATCH OUT

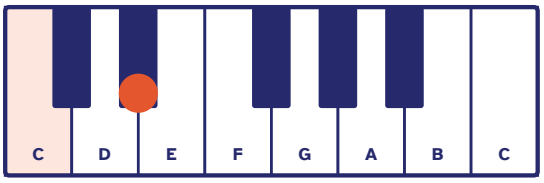
- × Going the wrong way round: clockwise adds sharps, anticlockwise adds flats.
- × Forgetting the relative minors on the middle ring.

C major's neighbours

KEY	WHERE ON THE CIRCLE	WHY IT MATTERS
G major	one step clockwise	the dominant: the strongest pull home
F major	one step anticlockwise	the subdominant: a gentle move away
A minor	straight in from C	the relative minor: the same notes
E minor, D minor	in from G and F	the relatives of the neighbours

IN ACTION

Find a relative minor on the circle



1. **Find the major key** on the outer ring: E♭.
2. **Look straight in** to the middle ring: c.
3. **Check:** three semitones down from E♭ is C.

The relative minor of E♭ major is **C minor**, the same three flats.

SEE ALSO

Keys and key signatures p. 40 · Finding the key p. 42 · Chords: Harmony (later)

PRACTISE IT

Workbook Pitch 5 · Online: Scale finder, Library › Related keys

PART I · PITCH · TIPS

When a note sounds wrong

Most wrong notes come from a handful of slips. Name the slip first, then fix only that note: playing the whole piece again rarely finds it.
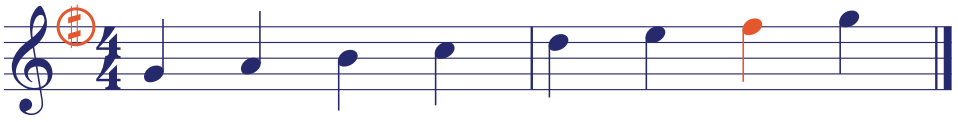
TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
A note sounds sour, though you read it right	**the key signature**: is this note sharp or flat all the way through?	p. 40
A sharp or flat vanishes, or stays too long	**the bar line**: an accidental lasts only to the end of its bar	p. 30
The left hand reads like the right	**the clef**: bass lines are G B D F A	p. 26
You land an octave too high or too low	**a landmark**: middle C, treble G or bass F	p. 26
Notes above or below the staff slow you down	**the ledger lines**: count on from the nearest landmark	p. 24
A leap lands on the wrong key	**the interval**: count the letters, then see the shape	p. 32
A minor piece sounds odd near a cadence	**the raised 7th** of the harmonic minor	p. 38

IN ACTION

Find the slip



1. **Read the key signature:** one sharp, on the top line, F (ringed).
2. **The red note sits on that line**, so the sharp belongs to it too.
3. **No natural cancels it**, so this F is F♯, here and all through the piece.

Play **F♯**, the black key. Plain F is the most common slip in G major.

HOW TO

Three checks before you play

1. **The clefs:** which staff is treble, which is bass?
2. **The key signature:** say its sharps or flats aloud, then find them in the first line.
3. **The landmarks:** find middle C, treble G and bass F, and read the other notes from them.

THINK OF IT Like proofreading: a spelling mistake jumps out once you know which words you usually get wrong. Keep a short list of your own slips, and check those first.

PART II · WHEN, AND FOR HOW LONG

Time

Music moves through time. Underneath it runs a steady **pulse**; the pulse is grouped into **bars**; and every note and rest takes a measured share of it. This part goes from that big picture down to the length of a single note.

Pulse and bar — the steady beat, in groups of 2, 3 or 4 › **Note values** — how long each sound and silence lasts › **Longer notes** — dots and ties › **Time signature** — how many beats, and which note is one beat › **Rhythm** — long and short, on and off the beat

IN THIS PART

Pulse, tempo and the bar 48
Notes and rests 50
How long is a note? 52
Time signatures 54
Compound and irregular time 56
Rhythm: groups and upbeats 58
Rhythm: off the beat 60
Tips: when the rhythm slips 62

PART II · TIME · 1

Pulse, tempo and the bar

Under almost every piece runs a **pulse**: a steady beat, like a heartbeat or a clock. Its speed is the **tempo**. Beats are grouped into **bars**, and the first beat of every bar is the strongest.

Tempo

Tempo is measured in beats per minute. A **metronome mark** at the top of the music shows it: ♩ **= 60** means 60 crotchet beats a minute, one beat a second. ♩ **= 120** is twice as fast. Tempo words such as *Andante* and *Allegro* come with Expression, later in the book.

Strong and weak beats

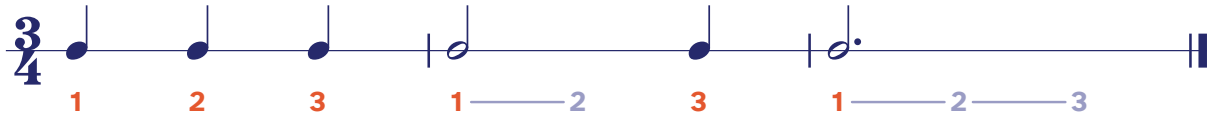
BEATS	PATTERN	FEELS LIKE
2	**Strong** weak	a march
3	**Strong** weak weak	a waltz
4	**Strong** weak **medium** weak	most songs

Bars and bar lines

SIGN	NAME	MEANING
	bar line	divides the music into bars
	double bar line	marks the end of a section, or a change of key, time signature or tempo
	final bar line	marks the end of the piece
	repeat signs	play the passage between them again; if there is no start sign, go back to the beginning

THINK OF IT The beat is the stage: it is there before anyone walks on. The notes are the actors who come and go on it, and the bar lines divide the stage into scenes.

Counting



Three bars of 3/4. Every beat is counted, even inside a long note: the red number is where you play, the grey numbers are the beats you hold.

Count aloud at one speed, a little louder on beat 1. With a metronome, start slower than you think and speed up only when the passage is steady.

WATCH OUT

- × Speeding up in the easy bars and slowing down in the hard ones: choose a tempo you can keep in the hardest bar.
- × Counting the notes instead of the beats.

PART II · TIME · 1 · CONTINUED

IN ACTION

Is this a march or a waltz?
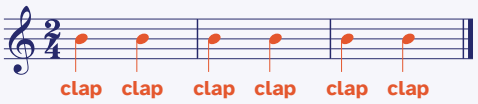


1. **Tap a steady beat** as you clap or listen.
2. **Find the strong one:** the red beats feel heavier.
3. **Count the gap** between strong beats: 1 2 3, 1 2 3.

Groups of three: a **waltz**, in 3/4. Groups of two would be a march.

THINK OF IT The pulse is a heartbeat or a ticking clock: it keeps going even when nothing is played.

YOU KNOW THIS

Radetzky March (Johann Strauss I)



At the New Year concert the audience claps along on **every beat**. That steady clap is the pulse; the orchestra's tune moves freely over it.

Try it. Put on any march and clap only the beat, not the tune.

IN ACTION

Feel the beat before you play



1. **Read the mark:** ♩ **= 96**, 96 crotchet beats a minute.
2. **Listen first:** set the metronome and let one full bar go by, counting "1 2 3" in your head.
3. **Step in** on the next beat 1, the way you step onto a moving escalator.

The first note lands **on the beat**, not a guess.

THINK OF IT To take the plate you want off a sushi train, or to step onto an escalator, you watch the movement first, then move with it at the right moment. Music is already moving before your first note: feel the beat for a bar, then step in.

WATCH OUT

- × Speeding up in the easy bars and slowing down in the hard ones: the pulse stays the same all the way through.
- × Counting only the notes you play: the beat goes on through long notes and rests.

SEE ALSO
Time signatures p. 54 · Tempo words: Expression (later)

PRACTISE IT
Workbook Level 0 · Pulse · Online: Metronome

PART II · TIME · 2

Notes and rests

A note's shape shows how long it lasts: each plain note lasts **half as long** as the one above it. Every note has a **rest** of the same length, a silence counted just as carefully as a sound.

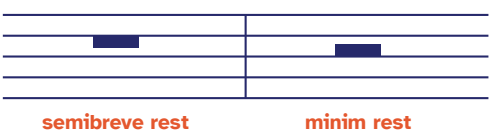
NOTE	REST	NAME	ALSO CALLED	BEATS*	FIRST MET
		breve	double whole note	**8**	Level 4
		dotted semibreve	dotted whole note	**6**	*Extra*
		semibreve	whole note	**4**	Level 0
		**dotted minim**	dotted half note	**3**	Level 0
		**minim**	half note	**2**	Level 0
		**dotted crotchet**	dotted quarter note	**1½**	Level ½
		**crotchet**	quarter note	**1**	Level 0
		**dotted quaver**	dotted eighth note	**¾**	Level 3
		**quaver**	eighth note	**½**	Level 0
		**semiquaver**	sixteenth note	**¼**	Level 2
		**demisemiquaver**	32nd note	**⅛**	*Extra*
		**hemidemisemiquaver**	64th note	**1/16**	*Extra*

* With the crotchet as one beat, as in 2/4, 3/4 and 4/4. *Extra*: beyond Level 8.

Parts of a note

The **head** sits on a line or in a space. A **stem** goes up on the right of the head or down on the left. **Flags** on the stem halve the value each time; groups of quavers are joined by **beams** instead of flags.

Which rest is which?



The semibreve rest **hangs** below the fourth line; the minim rest **sits** on the middle line.

PART II · TIME · 2 · CONTINUED

Why whole, half and quarter?

Think of each beat as a pizza. A bar of 4/4 is four pizzas. The note that fills the whole bar is the **whole note**; the others are a half, a quarter, an eighth and a sixteenth of it. Two quavers beamed together fill one pizza. A rest is the same share left empty: shaded with lines.

semibreve	whole note	4 pizzas · 4 beats	
semibreve rest	silence	4 empty pizzas · 4 beats	
minim	half note	2 pizzas · 2 beats	
minim rest	silence	2 empty pizzas · 2 beats	
crotchet	quarter note	1 pizza · 1 beat	
crotchet rest	silence	1 empty pizza · 1 beat	
quaver	eighth note	half a pizza · ½ beat	
quaver rest	silence	half an empty pizza · ½ beat	
semiquaver	sixteenth note	a quarter of a pizza · ¼ beat	
semiquaver rest	silence	a quarter of an empty pizza · ¼ beat	

IN ACTION

How long is each sound, and each silence?



1. **Name each shape:** minim, crotchet, crotchet rest.
2. **Count its beats:** 2, 1 and 1.
3. **Count the rests too:** the red rests last as long as their notes.

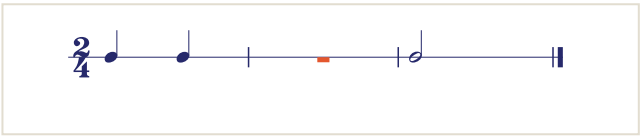
Every bar adds up to **4 beats**, silences included.

THE SEMIBREVE REST'S SPECIAL POWER

Most rests last a fixed number of beats. The **semibreve rest** can also mean **a whole bar of silence** in any time signature: 4 beats in 4/4, but 3 in 3/4 and 2 in 2/4. It is then centred in the bar. No other rest can do this.

One more rule: keep the middle of a 4/4 bar visible. Silence on beats 2–3 is two crotchet rests, never one minim rest.





SEE ALSO

Pulse and the bar p. 48 · How long is a note? p. 52

PRACTISE IT

Workbook Level 0 · Note values · Online: Library › Notes and rests
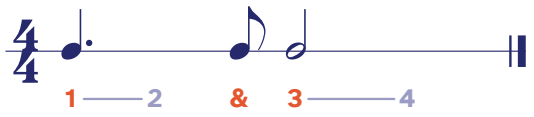
PART II · TIME · 3

How long is a note?

Notes and rests have fixed lengths (p. 50). Two signs make a note longer: the **dot**, which adds half, and the **tie**, which joins two notes into one sound.

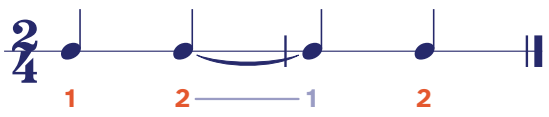
The dot

A dot adds **half the note's value**: a dotted minim lasts 2 + 1 = 3 beats; a dotted crotchet 1 + ½ = 1½. A **double dot** adds half, then a quarter: a double-dotted minim lasts 2 + 1 + ½ = 3½.



The tie

A curve joining two notes **of the same pitch** makes one longer sound: play the first, hold through the second. Ties can cross a bar line, where a dot cannot.
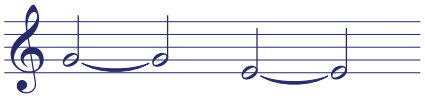


The dot, in pizzas

The dot is a helper that adds **half of whatever its note is worth**. Dark slices: the note. Lighter: what the dot adds. Lightest: a second dot, half of the first dot.

dotted minim		2 + **1** (half of 2) = **3 beats**
dotted crotchet		1 + **½** (half of 1) = **1½ beats**
dotted quaver		½ + **¼** (half of ½) = **¾ beat**
double-dotted minim		2 + **1** + ½ = **3½ beats**

Tie or slur?



Tie: the same pitch. Play once and hold: G for 4 beats, then E for 4.



Slur: different pitches. Play every note, smoothly joined (legato).

IN ACTION

Which is longer: a dotted minim or a minim tied to a quaver?



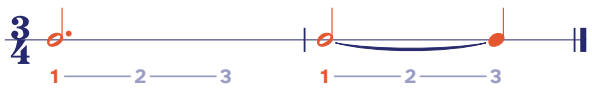
1. **Dotted minim:** 2 + 1 = 3 pizzas.
2. **Minim tied to a quaver:** 2 + ½ = 2½ pizzas.
3. **Compare:** 3 is half a beat more than 2½.

The **dotted minim** is longer, by half a beat.

PART II · TIME · 3 · CONTINUED

IN ACTION

Dot or tie: which is longer?

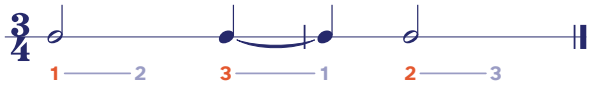


1. **The dotted minim:** 2 + half of 2 = 3 beats.
2. **The tied notes:** a minim tied to a crotchet, 2 + 1 = 3 beats.
3. **Play both:** one sound, held for three beats.

They are **the same length**. A tie can also cross a bar line, which a dot cannot.

IN ACTION

Count through a tie across the bar line
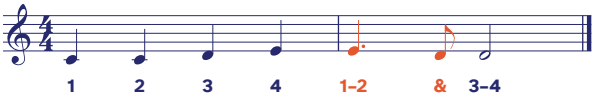


1. **Bar 1:** minim on 1–2, crotchet on 3.
2. **The tie** joins beat 3 to beat 1 of the next bar: one sound.
3. **Keep counting:** 3, 1 held, then 2–3 on the minim.

A tie can cross the bar line: **play once, count on**.

IN ACTION

A dotted crotchet you already know: Ode to Joy



1. **Bars 3–4 of Beethoven's Ode to Joy.** Sing it: "C C D E, E — D D".
2. **The red E is a dotted crotchet:** 1 + ½ = 1½ beats, held through "1 & 2".
3. **The quaver D** falls on the "&" of beat 2; the last D takes beats 3 and 4.

Long–short, then long: the dot is why the tune **leans** before it lands.

IN ACTION

Tie or slur? Read the curves



1. **The first curve** joins two Gs: the same pitch, so a **tie**.
2. **Play G once** and hold it for both minims: 4 beats.
3. **The second curve** covers C B A G: different pitches, so a **slur**: play each, smoothly joined.

Same pitch: **tie** (one sound). Different pitches: **slur** (legato).

SEE ALSO

Time signatures p. 54 · Rhythm p. 58

PRACTISE IT

Workbook Level 0 · Note values · Online: Library › Notes and rests

PART II · TIME · 4

Time signatures

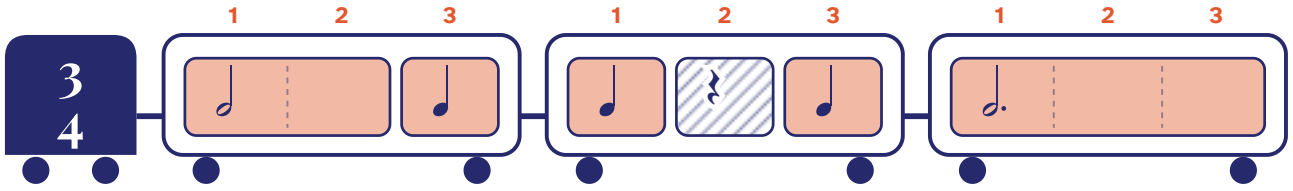
Two numbers at the start of the music. The **top number** says how many counts are in each bar; the **bottom number** says which note gets one count.

The bottom number

BOTTOM	NOTE	ONE COUNT IS A	FIRST MET
2	𝅗𝅥	minim	Level 3
4	♩	crotchet	Level 0
8	♪	quaver	Level 1
16	𝅘𝅥𝅯	semiquaver	*Extra*

Think of a train

In 3/4 every carriage (bar) has exactly **3 seats**; each seat is one crotchet beat. Notes book seats: a minim takes 2, a crotchet 1. A rest books a seat and leaves it **empty** (shaded). Every seat in every carriage must be booked.



minim + crotchet · crotchet, crotchet rest, crotchet · dotted minim: 3 seats each

Simple time: each beat divides in two

TIME SIGNATURE	EACH BAR HOLDS	FIRST MET
2/4	2 crotchets	Level 0
3/4	3 crotchets	Level 0
4/4 𝄴	4 crotchets (the C sign: common time)	Level 0
2/2 𝄵	2 minims (the cut C: cut common time)	Level 3
3/2	3 minims	Level 3
4/2	4 minims	*Extra*
3/8	3 quavers	Level 3

PART II · TIME · 4 · CONTINUED

Changing time signature

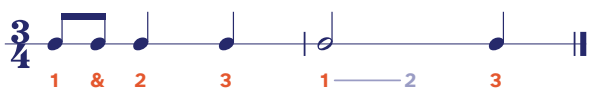
A new time signature can appear in the middle of a piece, right after a bar line; it holds until the next one. A time signature is written only once, not on every line.

WATCH OUT

- × Reading 6/8 as six beats: at most speeds it moves in two.
- × Mixing up 3/4 and 6/8: both hold six quavers, grouped differently (p. 56).

IN ACTION

Read a time signature
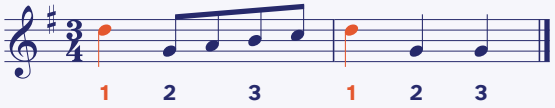


1. **Top number 3:** three counts in each bar.
2. **Bottom number 4:** a crotchet is one count.
3. **Check a bar:** two quavers + crotchet + crotchet = 3.

3/4: **three crotchet beats** in every bar.

YOU KNOW THIS

Minuet in G (from the Notebook for Anna Magdalena Bach)



Every bar holds **three crotchet beats**: 1 2 3, 1 2 3, the step of the dance. That is 3/4.

Try it. Count 1 2 3 aloud while someone plays it.

IN ACTION

Which time signature fits?



1. **Find the strong beats** by clapping, accenting the first note of each bar.
2. **Count each bar:** two crotchets, or one minim: 2 beats.
3. **Each beat is a crotchet.**

It is in **2/4**: two crotchet beats a bar.

SEE ALSO

Pulse and the bar p. 48 · Compound time p. 56

PRACTISE IT

Workbook Level 0 · Time signatures · Online: Library › Time signatures

PART II · TIME · 5

Compound and irregular time

Compound time has dotted beats that divide into three; irregular time has five or seven counts, grouped unevenly. (9/4, 12/4 and the /16 signatures exist but are rare.)

3/4 or 6/8?



3/4: six quavers in three twos



6/8: six quavers in two threes

SIMPLE OR COMPOUND?

In **simple time** each beat divides into **two**. In **compound time** each beat is a dotted note and divides into **three**.

Quick test: a top number of 6, 9 or 12 means compound. Divide it by 3 to find the beats: 6/8 has 2, 9/8 has 3, 12/8 has 4.

Compound time

TIME SIGNATURE	EACH BAR HOLDS	FIRST MET
6/8	2 dotted crotchets	Level 1
9/8	3 dotted crotchets	Level 3
12/8	4 dotted crotchets	Level 3
6/4	2 dotted minims	Level 4

Counting 6/8



Slowly, count all six quavers. At speed, feel two beats, on 1 and 4: “1 & a, 2 & a”.

THINK OF IT Like a skipping rope: two big swings, each with three little steps inside.

PART II · TIME · 5 · CONTINUED

Irregular time

TIME SIGNATURE	EACH BAR HOLDS	FIRST MET
5/4	5 crotchets: 3 + 2 or 2 + 3	Level 5
7/4	7 crotchets: 4 + 3 or 3 + 4	*Extra*
5/8	5 quavers: 3 + 2 or 2 + 3	*Extra*
7/8	7 quavers: 2 + 2 + 3 or 3 + 2 + 2	*Extra*

UNEVEN GROUPS IN 4/4

Many dances keep 4/4 but group the eight quavers 3 + 3 + 2: "1 2 3, 1 2 3, 1 2".



IN ACTION

Count 6/8 in two
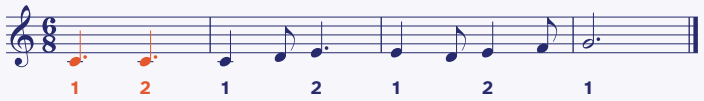


1. **Slowly, count all six:** 1 2 3 4 5 6.
2. **Lean on 1 and 4**, the red notes.
3. **Faster, count only those:** two beats, each a dotted crotchet.

6/8 has **two beats** in a bar, each split into three.

YOU KNOW THIS

Row, Row, Row Your Boat

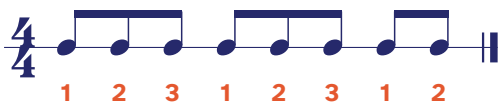


Two big beats in every bar, each a **dotted crotchet**: "Row, row". Inside each beat, "gently" splits it into three. That is 6/8.

Try it. Sway twice per bar while you sing it.

IN ACTION

Hear 3 + 3 + 2 inside 4/4



1. **Count eight quavers** in a bar of 4/4.
2. **Group them** 3 + 3 + 2, accenting 1, 4 and 7.
3. **Keep the beat** in your foot: it stays in four.

The bar is still **4/4**; only the accents are uneven.

SEE ALSO

Time signatures p. 54 · Rhythm: groups and upbeats p. 58

PRACTISE IT

Workbook Levels 1–4 · Compound time · Online: Rhythm clock
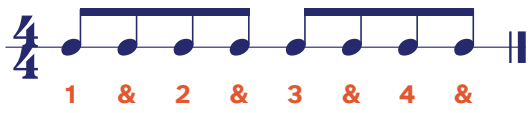
PART II · TIME · 6

Rhythm: groups and upbeats

Rhythm is the pattern of long and short sounds over the pulse. The pulse stays steady; the rhythm moves around it. The way notes are grouped on the page shows where the beats are.

Grouping with beams

Beams join the notes of **one beat**, so the beats stay visible. In 4/4 a beam may cover beats 1–2 or 3–4, but never cross the middle of the bar (except in a deliberate grouping such as 3 + 3 + 2). In compound time, quavers are beamed in threes.



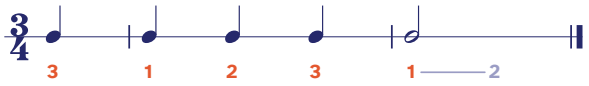
✓ The middle of the bar can be seen.



× A beam across beats 2–3 hides it.

The upbeat (anacrusis)

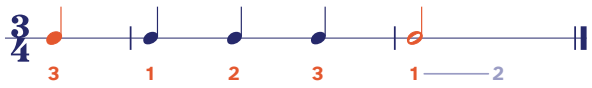
A piece can begin **before** the first full bar, with one or more notes called an upbeat or anacrusis. The last bar is then short by the same amount, so the two together make one full bar. Count the missing beats in silence before you start.



Count “1 2” silently, then play on 3.

IN ACTION

Balance the upbeat

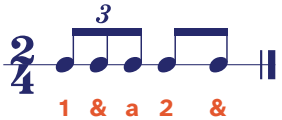


1. **The upbeat:** one beat (count 3) before the first full bar.
2. **The last bar:** only two beats.
3. **Add them:** 1 + 2 = 3, one whole bar of 3/4.
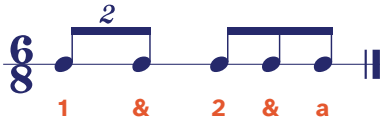
The upbeat and the last bar **make one full bar** together.

THINK OF IT An upbeat is the breath before you speak: “and-ONE”. You lift on the upbeat so the strong beat can land.

Triplets and duplets



Triplet: three exactly equal notes in the time of two. First met: Level 2.



Duplet: two notes in the time of three, in compound time. First met: Level 2.

PART II · TIME · 6 · CONTINUED



Two quavers: one beat cut in 2



A **triplet**: one beat cut in 3



Four semiquavers: one beat cut in 4
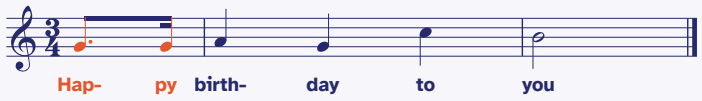
Counting smaller values



Crotchet, two quavers, four semiquavers and a triplet (“1 & a”): one beat each.

YOU KNOW THIS

Happy Birthday
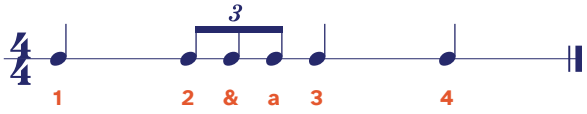


“Hap-py” comes **before** the first strong beat: an upbeat. The strong beat lands on “birth”.

Try it. Sing it and clap only on “birth”, “day” and “you”: you feel where bar 1 starts.

IN ACTION

Fit a triplet into a bar



1. **Count the bar:** four crotchet beats, “1 2 3 4”.
2. **Find the 3:** on beat 2, three quavers share one beat.
3. **Say it:** “1, 2 & a, 3, 4”, keeping beats 3 and 4 steady.

The triplet squeezes three notes into **beat 2**; the beat itself never moves.

IN ACTION

Count four semiquavers



1. **One beat** cut into four equal notes.
2. **Say it:** 1 e & a, evenly.
3. **The next beat** starts again on 2.

Four semiquavers fill **one beat**: a quarter each.

SEE ALSO
Notes and rests p. 50 · Rhythm: off the beat p. 60

PRACTISE IT
Workbook Levels ½–4 · Rhythm · Online: Rhythm clock

PART II · TIME · 7

Rhythm: off the beat

Once the pulse is secure, rhythm can pull against it. Every one of these effects needs a steady beat underneath.

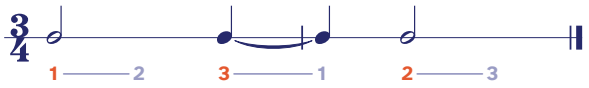
Syncopation



The crotchet starts on “&” and is held through beat 2.

Level 4. Feel beat 2 even though nothing is played on it.

Hemiola



Two bars of 3/4 felt as three minims.

Level 4. Common at cadences in Baroque dances.

Swing Level ½

Common from the first pieces: jazz, blues and many easy modern pieces are marked *swing*. Pairs of quavers are written evenly but played **long–short**, like the first and last notes of a triplet.

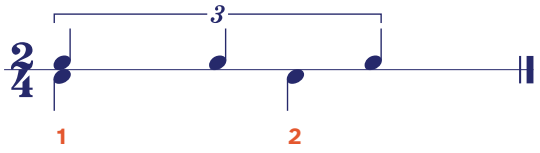


Written



Played

Polyrhythm Level 7



Three against two: a crotchet triplet above, two crotchets below, starting together.

Rubato Extra

Time is stretched and then given back, for expression: the music leans into a phrase and catches up after it. The pulse always returns. Marked *rubato*, or simply part of the style in much Romantic music.

WATCH OUT

- × Rushing the short notes after a long one: keep counting inside the long note.
- × Playing a triplet unevenly: three equal notes, not two quick and one long.
- × Letting syncopation drag the pulse: tap the beat with your foot.

THINK OF IT Syncopation is a step that lands between the beats, like a skip in your walk.

PART II · TIME · 7 · CONTINUED

IN ACTION

Count a syncopation



1. **Tap the beats:** 1, 2.
2. **Find the red crotchet:** it starts on the "&", between the beats.
3. **Hold it through beat 2**, then play bar 2 on the beats.

The accent falls **off the beat**: that is syncopation.

YOU KNOW THIS

The Entertainer (Scott Joplin)
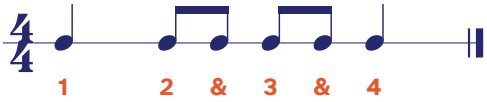


The high **C** keeps landing just after the beat, not on it: syncopation, the bounce of ragtime.

Try it. Tap the beat with your foot and hum the tune: hear the C slip between the taps.

IN ACTION

Swing this bar

Swing

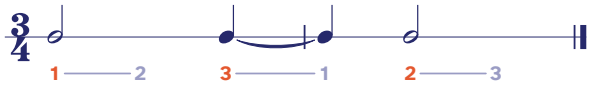


1. **Find the quaver pairs:** beats 2 and 3.
2. **Play each pair long–short:** "doo-ba, doo-ba".
3. **Leave the crotchets alone:** beats 1 and 4 stay as written.

Only the **pairs** swing; the beat underneath stays even.

IN ACTION

Count a hemiola



1. **Two bars of 3/4:** six crotchet beats.
2. **Regroup them** in twos: 1 2 | 3 4 | 5 6.
3. **Feel three big beats** across the two bars.

Two bars of three become **three bars of two** for a moment: a hemiola.

SEE ALSO

Rhythm: groups and upbeats p. 58 · Words for how to play: Expression (later)

PRACTISE IT

Workbook Levels ½–4 · Rhythm · Online: Rhythm clock

PART II · TIME · TIPS

When the rhythm slips

A rhythm that slips is nearly always a beat that was not counted. Find which beat, count it out loud, and the rest of the bar usually falls into place.

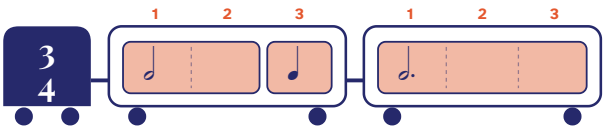
TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
It speeds up or slows down	**the pulse**: count one bar aloud before you start	p. 48
A rest gets skipped	**the rest**: it is counted just like a note	p. 50
Long notes come out short	**the hold**: count every beat of a minim, dot or tie	p. 52
The bar feels too long or too short	**the time signature**: how many counts, which note is one	p. 54
6/8 sounds like 3/4	**the dotted beat**: two beats, each split in three	p. 56
You lose your place in fast notes	**the beams**: each group is one beat	p. 58
An off-beat note lands on the beat	**the "&"**: keep the beat in your foot and count it	p. 60

IN ACTION

Count the long notes



1. **Say 1 2 3** once through before you play, to feel the beat.
2. **The minim books seats 1 and 2:** keep saying "2" while it sounds.
3. **The dotted minim books all three:** let go on the next "1", not before.

Long notes are where counting slips: **say every seat** they book.

Four ways to steady a rhythm

1. **Count out loud**
   Say the counts while you play, not in your head: a slip you can hear is a slip you can fix.

2. **Tap, then play**
   Tap the rhythm on the lid and say the counts; play only when the tapping is steady.

3. **Walk the beat**
   Step the pulse while you clap the rhythm, so the beat lives in your feet.

4. **Split the beat**
   For dots and quavers, count "1 & 2 &": the extra "&" shows exactly where each note goes.

THINK OF IT Like a dance partner: the beat leads and the rhythm follows. Lose the beat and every step after it goes wrong.

PART III · WHICH NOTES SOUND TOGETHER

Harmony

A melody moves one note at a time. Harmony is what sounds with it: the chords underneath. This part starts with the triad, three notes stacked in 3rds, and ends with the chords that carry a melody, close a phrase and pull a piece home.

Triads three notes stacked in 3rds › **Kinds** major, minor, diminished, augmented › **Chords in a key** I, IV and V › **Inversions** the same chord, a new bass › **Cadences** the endings of phrases

IN THIS PART

Chords and triads 64
Four kinds of triad 66
Chords in a key 68
Inversions 70
Cadences 72
Chords under a melody 74
The dominant 7th 76
Tips: when a chord sounds wrong 78

PART III · HARMONY · 1

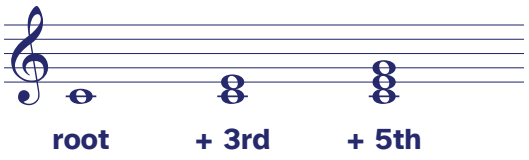
Chords and triads

A **chord** is two or more notes sounding together. The most common chord is the **triad**: three notes stacked in 3rds. When the notes are stacked in 3rds, the bottom note is the **root**, and it gives the chord its name.

Root, 3rd and 5th

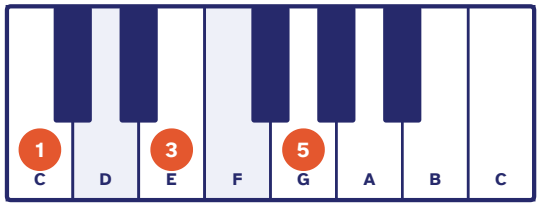
Start on any note: that is the **root**. Add the note a **3rd** above it, then the note a **5th** above the root. On the staff each note sits on the next line up, or in the next space up. C, E and G make the triad of C.

First met: Level ½ (the tonic triad of C, G and F major).



The root, then a 3rd above it, then a 5th above the root.

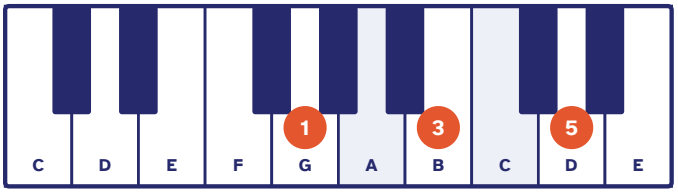
Play one, skip one



On the white keys a triad is **play one, skip one, play one, skip one, play one**: C (skip D) E (skip F) G. The red keys are the root, the 3rd and the 5th; the skipped keys are shaded.

IN ACTION

Build a triad on G



1. **Start on G:** the root.
2. **Skip A, play B:** the 3rd.
3. **Skip C, play D:** the 5th.

G B D: the triad of **G**. Same shape as C E G, just moved up.



THINK OF IT A snowman: three balls stacked, each one on the next, with one note skipped between them. The bottom ball, the root, holds it up and gives it its name.

All lines or all spaces

A stacked triad sits on **three lines in a row** or in **three spaces in a row**, in a neat stack. If one note breaks the pattern, it is not a triad.



Lines, spaces, lines, spaces: every one a triad.

PART III · HARMONY · 1 · CONTINUED

Solid and broken

Played together, a chord is **solid** (or blocked). Played one note at a time, it is **broken**. Spread up and down the keyboard, a broken chord becomes an **arpeggio**. Left-hand accompaniments, the arpeggios in technical work and the chord names over a song are all triads.

First met: Level 0; arpeggios from Level 1.



The same three notes, solid then broken.

IN ACTION

Is it a triad?
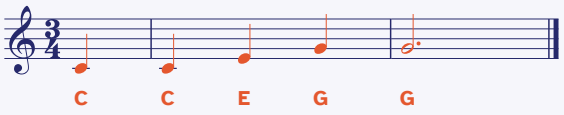


1. **Chord 1:** D F A, all in spaces, each a 3rd apart: **a triad**.
2. **Chord 2:** D E A: D to E is a 2nd. **Not a triad.**
3. **Chord 3:** D G C, stacked in 4ths. **Not a triad.**

Only chord 1 is a triad. Check that each note is a **3rd** above the one below.

YOU KNOW THIS

The Blue Danube (Johann Strauss II), here in C



The most famous waltz tune simply climbs the **C triad**: **C, C E G, G**. Root, 3rd, 5th, then it holds the top: that is why it sounds so open.

Try it. Play C E G one at a time, then hum the waltz over it.

WATCH OUT

- × Stacking 2nds or 4ths: a triad skips exactly one letter each time.
- × Naming a chord from its top note: the root names it (at the bottom when the notes are stacked in 3rds).

SEE ALSO
Counting intervals p. 32 · Major scales p. 36

PRACTISE IT
Workbook Harmony 1 · Online: Library › Chords

PART III · HARMONY · 2

Four kinds of triad

Every triad is two 3rds stacked. To tell the kinds apart, count the keys **in between**, black and white, that you do not play: root to 3rd, then 3rd to 5th. Those two numbers name the kind, and its sound.

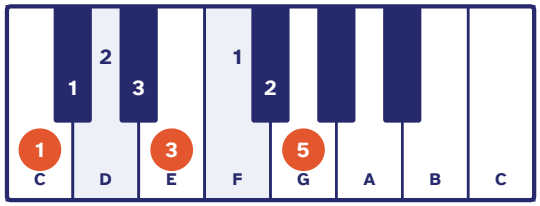
KIND	KEYS BETWEEN	ON C	CHORD NAME	FIRST MET
major	**3 + 2**	C E G	C	Level ½
minor	**2 + 3**	C E♭ G	Cm	Level 2
diminished	**2 + 2**	C E♭ G♭	C° or Cdim	Level 5
augmented	**3 + 3**	C E G♯	C+ or Caug	Level 6

Chord names: a letter alone means major (C); m means minor (Cm). You see them above the melody in songbooks and lead sheets: C | Am | F | G. In interval names, 3 keys between is a major 3rd and 2 is a minor 3rd (p. 34).



The four kinds on C. Only the 3rd and the 5th move; the root stays.

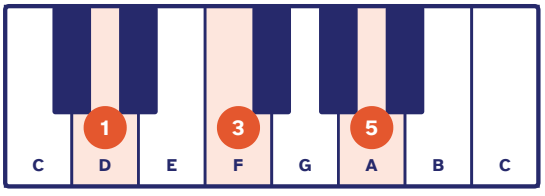
Count the keys in between



Play C and E and count the keys you skip, black and white: C♯, D, D♯ is **3**. Then E to G: F, F♯ is **2**. **3 + 2 is major.** Any root works the same way: from D, skip 3 to F♯, then 2 to A: D F♯ A, D major.

IN ACTION

Major or minor? D F A



1. **Root to 3rd:** D to F skips D♯ and E: **2**.
2. **3rd to 5th:** F to A skips F♯, G and G♯: **3**.
3. **2 + 3** is minor.

D F A is **D minor** (Dm). Three of the white-key triads are minor: D, E and A.

THINK OF IT Two floors of a building: a tall floor (3 keys between) and a short one (2). Major: tall, then short, 3 + 2. Minor: short, then tall, 2 + 3. Diminished: two short, 2 + 2. Augmented: two tall, 3 + 3.

PART III · HARMONY · 2 · CONTINUED

The seven white-key triads, sorted
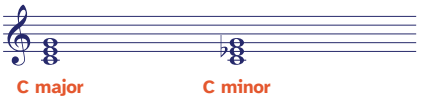


Always the same on the white keys: **three major** (C, F, G), **three minor** (D, E, A), **one diminished** (B), no augmented. In any major key, chords I, IV and V are the major ones (p. 68).

How they sound

KIND	SOUNDS	HEARD IN
major	bright, settled, open	most cheerful tunes, fanfares
minor	darker, softer, sadder	lullabies, laments, film tension
diminished	tense, unstable, wants to move	suspense, the chord before home
augmented	floating, dreamy, unsettled	dream scenes, a passing colour

IN ACTION

Change one note
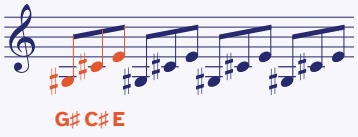


1. **Start with C major:** C E G.
2. **Lower the 3rd** one semitone: E becomes E♭.
3. **Keep the root and the 5th.**

C E♭ G: **C minor**. One semitone turns major into minor.

YOU KNOW THIS

Beethoven, Moonlight Sonata (first movement)



The famous slow triplets are the notes of a **C♯ minor triad**: C♯, E and G♯, rocking gently. Minor is why it sounds so dark and sad.

WATCH OUT

- × Counting the keys you play: count only the keys in between, black and white.
- × Forgetting that in diminished and augmented triads the 5th changes too.
- × Spelling with the wrong letter: a triad skips one letter, so C minor is C E♭ G, never C D♯ G.

SEE ALSO

Chords and triads p. 64 · Major, minor and perfect p. 34
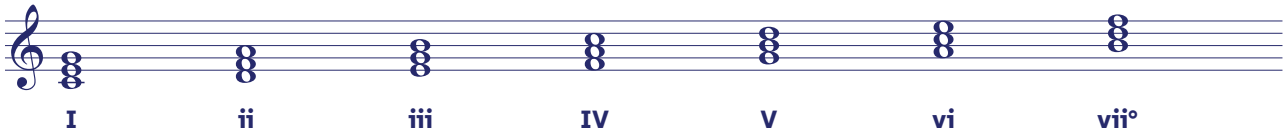
PRACTISE IT

Workbook Harmony 2 · Online: Library › Chords

PART III · HARMONY · 3

Chords in a key

Build a triad on every note of a scale and you have the chords of that key. **Roman numerals** name each one by its place in the scale: I on the 1st note, IV on the 4th, V on the 5th. Capitals are major, small numerals are minor, and small with ° is diminished.



The seven chords of C major. Only white keys, so the kinds come out as on p. 67.

NUMERAL	NAME	KIND	KEYS BETWEEN	IN C	IN G
I	tonic	major	3 + 2	C	G
ii	supertonic	minor	2 + 3	Dm	Am
iii	mediant	minor	2 + 3	Em	Bm
IV	subdominant	major	3 + 2	F	C
V	dominant	major	3 + 2	G	D
vi	submediant	minor	2 + 3	Am	Em
vii°	leading note	diminished	2 + 2	B°	F♯°

The same in every major key: I, IV and V are major; ii, iii and vi are minor; vii° is diminished. First met: I, IV and V at Levels 1–2; the other chords from about Level 4.

The three primary chords

I, IV and V are the strongest chords of a key. Between them they hold **every note of the scale**: C E G, F A C, G B D. That is why so many tunes need only these three.

THINK OF IT I is home. IV is a visit next door. V is the walk back, already looking at the front door: it wants to go home to I. Play C, F, G, C and hear it.

IN ACTION

Find I, IV and V in G major
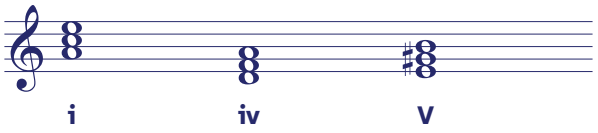


1. **Count up the scale:** G A B C D. The 1st note is G, the 4th C, the 5th D.
2. **Build a triad on each**, using the key signature: the F is F♯.
3. **Check with the keys between:** all three are 3 + 2, major.

In G major: **I = G**, **IV = C**, **V = D** (D F♯ A).

PART III · HARMONY · 3 · CONTINUED

In a minor key

In A minor the three main chords are **i** (A minor), **iv** (D minor) and **V**: E G♯ B, made major by the **raised 7th**, G♯, which pulls up to the home note A. Check: E G♯ B is 3 + 2, major.



A minor: i, iv and V. The G♯ makes V major.
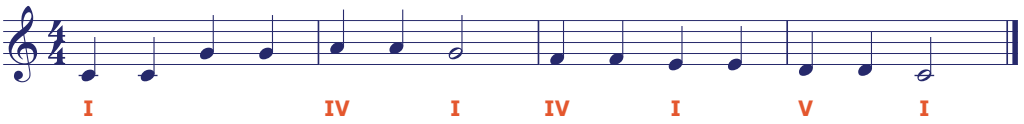
Which chord holds the note?

NOTE IN C	C	D	E	F	G	A	B
Chord	I or IV	V	I	IV	I or V	IV	V

The same idea works in every key: find which of I, IV and V holds the note.

IN ACTION

Choose a chord for each bar: Twinkle



1. **Look at the notes** in each bar (or half bar).
2. **Find I, IV or V** that holds them, from the table above.
3. **End V then I** (a perfect cadence, p. 72): the tune comes home.

Twinkle needs only **I, IV and V**: C, F and G.

YOU KNOW THIS

The twelve-bar blues

I C	I C	I C	I C
IV F	IV F	I C	I C
V G	IV F	I C	V G

Almost every blues, and a lot of rock 'n' roll, is just **I, IV and V** in twelve bars. Learn this chart and you can play along with hundreds of songs.

Try it. Play the chart in C with your left hand, one chord per bar, four beats each.

WATCH OUT

- × Writing every numeral big: ii, iii and vi are minor, so they are written small.
- × Forgetting the raised 7th in a minor key: V in A minor is E G♯ B, not E G B.

SEE ALSO

Four kinds of triad p. 66 · Major scales p. 36 · Cadences p. 72
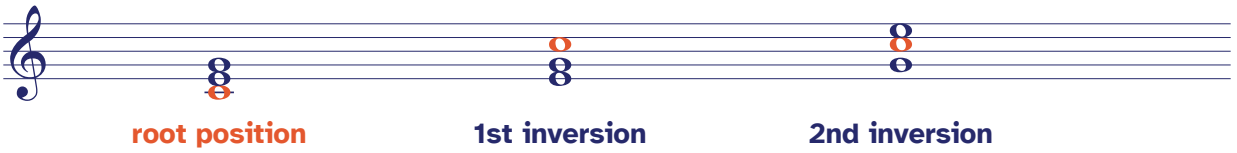
PRACTISE IT

Workbook Harmony 3 · Online: Library › Chords

PART III · HARMONY · 4

Inversions

The same three notes can be stacked in a different order. With the root at the bottom the chord is in **root position**. Move the root to the top and a new note is at the bottom: an **inversion**. The chord keeps its name; only the bass note changes.

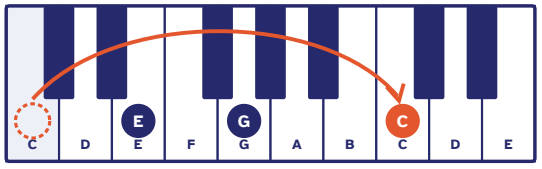


The red note is the root, C. Watch it climb: at the bottom, then on top, then in the middle. The chord is always C major.

POSITION	BASS NOTE	CHORD SYMBOL	NUMERAL	FIGURES	FIRST MET
root position	the root (C)	C	I	5/3	Level ½
1st inversion	the 3rd (E)	C/E	Ib	6/3 (6)	Level 3
2nd inversion	the 5th (G)	C/G	Ic	6/4	Level 5

Only the lowest note decides: E C G, or E G C E, is still a 1st inversion. Chord symbols write the bass after a slash: C/E is a C chord with E in the bass. Figures count the intervals above the bass, by letters: in E G C, G is a 3rd and C a 6th above E, so 6/3.

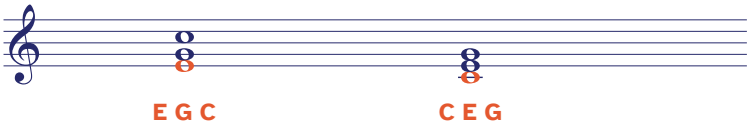
Roll the bottom note to the top



Play C E G, then lift the **root C** and put it on top (the red arrow): **E G C**. The E is now the bass: the 1st inversion. Do it again (G C E) for the 2nd inversion. In an inversion the stack has a **gap of a 4th** (G up to C): the note above the gap is the root.

IN ACTION

Name this chord: E G C



1. **Restack it in 3rds**, like the snowman: C E G.
2. **The bottom of the stack is the root:** C, so it is C major.
3. **Which note is in the bass?** E, the 3rd: 1st inversion.

E G C is **C major, 1st inversion**: C/E.

THINK OF IT The same three friends in a photo, swapping places: still the same group (the same chord), but whoever stands at the front (the bass) changes the picture.

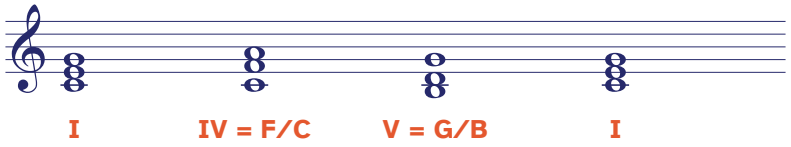
PART III · HARMONY · 4 · CONTINUED

Why inversions?

They let your hand **move as little as possible**: keep any note two chords share, and move the others to the nearest note of the next chord. Left-hand accompaniments and hymn parts are written this way.

IN ACTION

Play I, IV, V, I smoothly
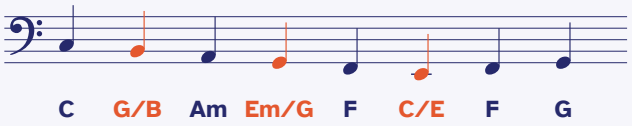


1. **I:** C E G. **IV:** keep C, move E up to F and G up to A: C F A.
2. **V:** no shared note, so each moves to the nearest: C to B, F to D, A to G: B D G.
3. **I again:** back to C E G. Each finger moved one or two keys.

IV is F/C and V is G/B: **inversions** keep the hand in one place.

YOU KNOW THIS

Pachelbel's Canon, the bass (here in C)



The famous bass walks **straight down the scale**: C B A G F E. It can, because some chords are **inversions**: G/B, Em/G and C/E put a different note in the bass.

Try it. Play the bass slowly and hum the famous violin tune over it.

Chords of C, every position

CHORD	ROOT POSITION	1ST INVERSION	2ND INVERSION
I (C)	C E G	E G C	G C E
IV (F)	F A C	A C F	C F A
V (G)	G B D	B D G	D G B
vi (Am)	A C E	C E A	E A C

WATCH OUT

- × Naming a chord from its bass note: in an inversion the bass is not the root. Restack it in 3rds first.
- × Thinking an inversion is a new chord: E G C is still C major.
- × Counting keys between before restacking: restack first (E G C → C E G), then count 3 + 2.

SEE ALSO

Chords and triads p. 64 · Chords in a key p. 68 · Cadences p. 72

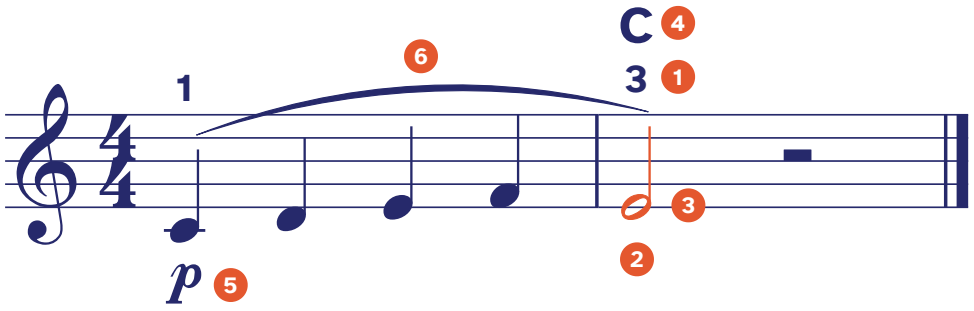
PRACTISE IT

Workbook Harmony 4 · Online: Library › Chords

PUTTING IT TOGETHER · 1

Everything one note tells you

Look at the red note. It carries six pieces of information at once, and each one can be read straight off the page. Read all six and the note is right; miss one and it is only nearly right.



1 The finger number

3: the middle finger plays it. Fingering is planned, not guessed.

PLAYING AND LISTENING (LATER)

2 The beat

Beat 1 of bar 2, and it is held for **two beats**.

PART II · TIME · P. 48

3 The note name

E: the white key just right of the two black keys.

PART I · PITCH · P. 22

4 The chord

The **C** above the staff: this E belongs to the C major chord.

PART III · HARMONY · P. 64

5 The marks

p and the slur: play it softly and smoothly.

EXPRESSION (LATER)

6 The phrase

The slur ends here, so the note lands gently, like the end of a sentence.

SHAPE AND STYLE (LATER)

THINK OF IT A note is like a word said aloud: its spelling, its timing, its tone of voice and its place in the sentence all change what it means.

THE SAME SIX, IN ANY MUSIC

A CLASSICAL SCORE
Most of it is written in: finger numbers, dynamics, slurs. Your job is to read every one.

A POP LEAD SHEET
A melody and chord names only. You choose the fingers, the marks and the shape yourself.

A JAZZ CHART
Chords and a style word such as Swing: the beat and the shape are yours to feel.

Try it on any piece: point to one note and find all six. Finger numbers count as much as the note name.

PUTTING IT TOGETHER · 2

When a passage won't come right

Name the problem first, then fix that one thing. Most hard bars are hard for one reason, and the six things a note tells you point straight to it.

TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
You stumble in the same place every time	**the finger number**	Playing and listening
It rushes, drags or the counting slips	**the beat**	Part II · Time
A note sounds wrong	**the note name**	Part I · Pitch
A chord sounds odd or thin	**the chord**	Harmony
It is correct but flat and dull	**the marks**	Expression
It does not seem to go anywhere	**the phrase**	Shape and style

Six ways to fix a hard bar

1. **Slow it right down**
   Slow enough to be right three times in a row, then a little faster.
2. **Hands separately**
   Each hand on its own until it is easy, then together, slowly.
3. **Start from the end**
   Play the last beat, then the last two, then three: you always finish on what you know best.
4. **Practise the join**
   The last note of one bar and the first of the next: most slips live there.
5. **Climb the metronome**
   Up four clicks each time it is clean; down eight if it falls apart.
6. **Put it back**
   Play two bars before and two after, so the fixed bar lives inside the music again.

Before you play a new piece: a thirty-second look

Key signature
Which sharps or flats, all the way through?

Time signature
How many beats in a bar, and which note is one beat?

Tempo and mood
The words at the top: how fast, what character?

Patterns
Bars that repeat or nearly repeat: learn them once.

THINK OF IT Like a doctor: name the symptom first, then treat that one thing. Practising everything harder rarely fixes the one bar that is wrong.